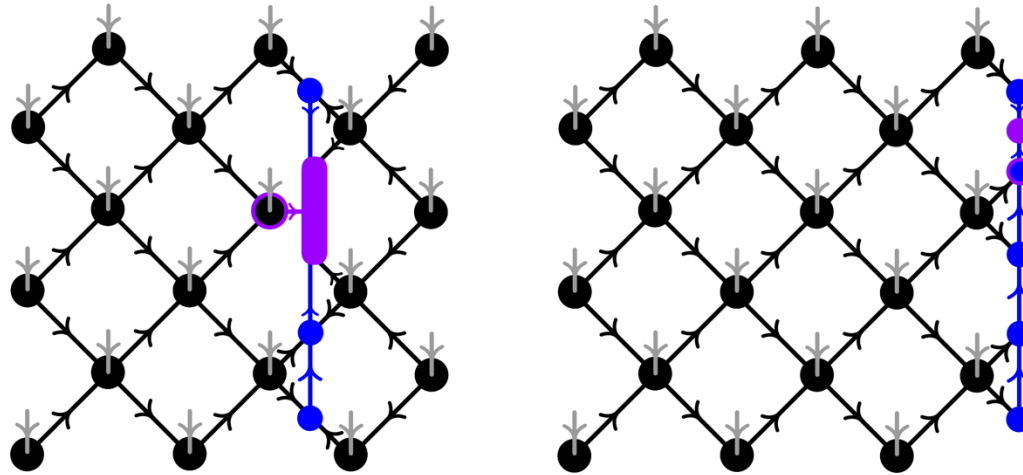


Variational Ground States and Quasiparticle Excitations in Isometric Tensor Network States

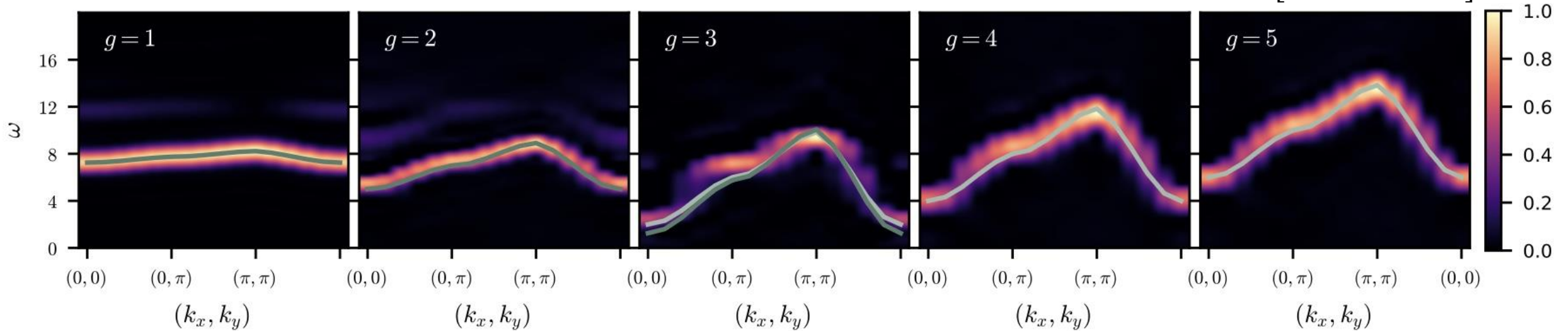
Lukas Julius Wittmann



Motivation: Dynamical structure factor

$$S(k, \omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt \sum_n e^{i(\omega t - k \cdot n)} C(n, t) \quad \text{with } C(n, t) = \langle \psi_0 | e^{iHt} \sigma_n^y e^{-iHt} \sigma_0^y | \psi_0 \rangle$$

[Lin et al. 2022]



1) From time evolution $C(n, t) = e^{iE_0 t} \langle \psi_0 | \sigma_n^y \underbrace{e^{-iHt} \sigma_0^y | \psi_0 \rangle}_{=|\psi(t)\rangle} \rangle$

2) Lowest branch of quasiparticle excitations $S(k, \omega) = \sum_{X,n} e^{-ik \cdot n} \langle \psi_0 | \sigma_n^y | \psi_X \rangle \langle \psi_X | \sigma_0^y | \psi_0 \rangle \delta(\omega - \epsilon_X)$

Transverse field Ising (TFI) model

$$H = -J \sum_{\langle n,m \rangle} \sigma_n^x \sigma_m^x - g \sum_n \sigma_n^z$$

Critical point

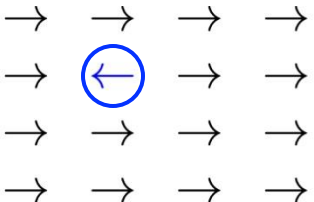
$g \ll J$ Ferromagnetic phase

$$|0\rangle = |\rightarrow \dots \rightarrow\rangle \text{ or } |0\rangle = |\leftarrow \dots \leftarrow\rangle$$

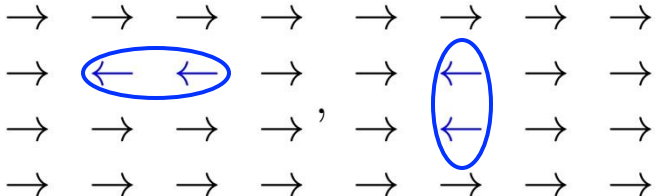
1D $|n\rangle = |\rightarrow \dots \rightarrow \underset{n}{\leftarrow} \dots \leftarrow\rangle$ with $\Delta_1 = 2J$

$$|n, m\rangle = |\rightarrow \dots \rightarrow \underset{n}{\leftarrow} \dots \underset{m}{\leftarrow} \rightarrow \dots \rightarrow\rangle \text{ with } \Delta_2 = 4J$$

2D



with $\Delta_1 = 8J$



with $\Delta_2 = 12J$

1D $g_c = J$
2D $g_c \approx 3.044J$

$g \gg J$ Paramagnetic phase

$$|0\rangle = |\uparrow \dots \uparrow\rangle$$

1D and 2D

$$|n\rangle = |\uparrow \dots \uparrow \underset{n}{\downarrow} \uparrow \dots \uparrow\rangle \text{ with } \Delta_1 = 2g$$

$$|n, m\rangle = |\uparrow \dots \uparrow \underset{n}{\downarrow} \uparrow \dots \uparrow \underset{m}{\downarrow} \uparrow \dots \uparrow\rangle \text{ with } \Delta_2 = 4g$$

Transverse field Ising (TFI) model

$$H = -J \sum_{\langle n,m \rangle} \sigma_n^x \sigma_m^x - g \sum_n \sigma_n^z$$

Critical point

$g \ll J$ Ferromagnetic phase

1D $g_c = J$

2D $g_c \approx 3.044J$

$g \gg J$ Paramagnetic phase

1D $|n\rangle = |\rightarrow \dots \rightarrow \underset{n}{\leftarrow} \dots \leftarrow\rangle$ with $\Delta_1 = 2J$

$\Rightarrow \mu = \Delta_1, t = g$

1D and 2D

$|n\rangle = |\uparrow \dots \uparrow \underset{n}{\downarrow} \uparrow \dots \uparrow\rangle$ with $\Delta_1 = 2g$

$\Rightarrow \mu = \Delta_1, t = J$

2D

$\begin{array}{cccc} \rightarrow & \rightarrow & \rightarrow & \rightarrow \\ \rightarrow & \circlearrowleft & \rightarrow & \rightarrow \\ \rightarrow & \rightarrow & \rightarrow & \rightarrow \\ \rightarrow & \rightarrow & \rightarrow & \rightarrow \end{array}$
 with $\Delta_1 = 8J$

$\Rightarrow \mu = \Delta_1 - g^2/(4J), t = g^2/(8J)$

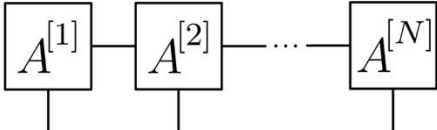
Perturbation theory $\Rightarrow H_{\text{eff}} = \mu \sum_n |n\rangle \langle n| - t \sum_{\langle n,m \rangle} (|n\rangle \langle m| + |m\rangle \langle n|)$

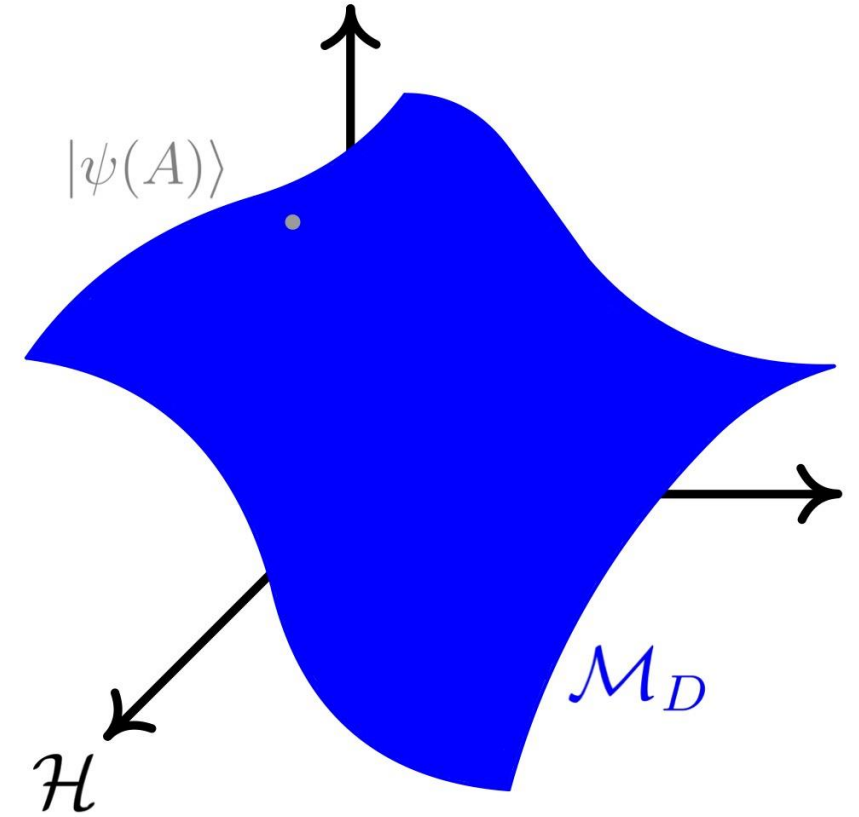
Variational Ground States and Quasiparticle Excitations in Isometric Tensor Network States

- 1) Matrix product states (MPS)
- 2) Isometric projected entangled pair states (isoPEPS) on diagonal square lattice
 - ⇒ New bulk-weighted boundary compression for DMRG²
 - ⇒ New quasiparticle excitation ansatz

Matrix product states (MPS)

Manifold

$$|\psi(A)\rangle = \boxed{A^{[1]}} - \boxed{A^{[2]}} - \dots - \boxed{A^{[N]}}$$
A diagram showing a sequence of three square boxes labeled $A^{[1]}$, $A^{[2]}$, and $A^{[N]}$ connected by horizontal lines. Each box has a vertical line extending downwards from its center.



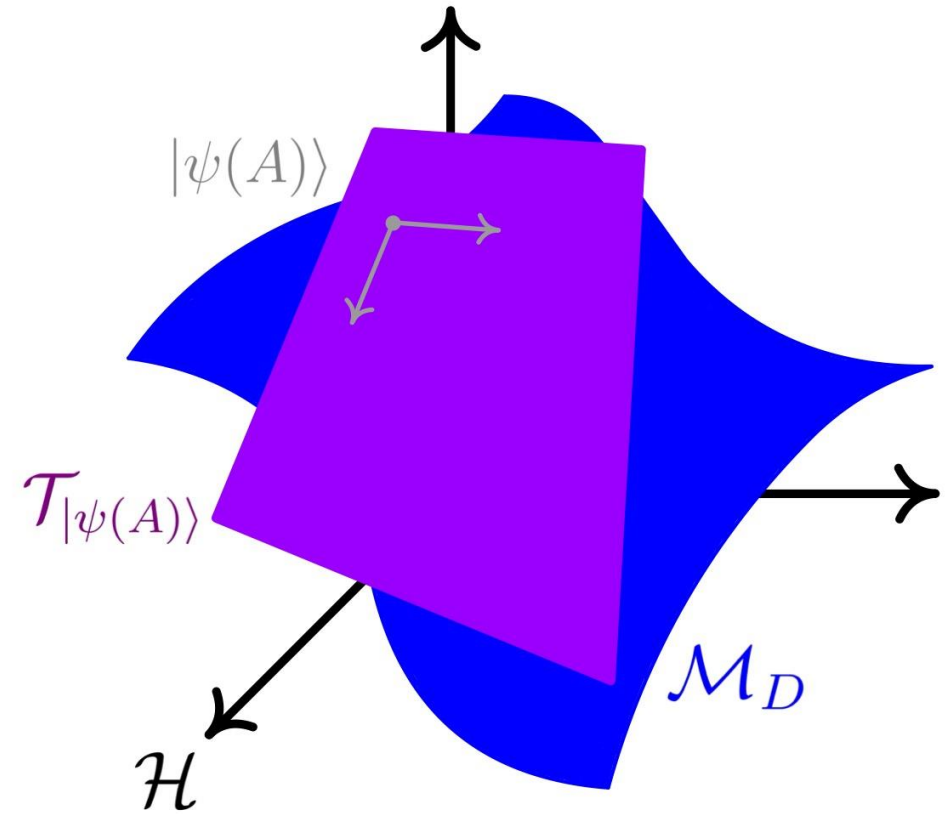
Matrix product states (MPS)

Manifold

$$|\psi(A)\rangle = \boxed{A^{[1]}} - \boxed{A^{[2]}} - \dots - \boxed{A^{[N]}}$$

Tangent space

$$\begin{aligned} |\psi(B; A)\rangle &= \sum_n \sum_{s, \alpha, \beta} B_{\alpha, \beta}^{[n]s} \left[\partial_{A_{\alpha, \beta}^{[n]s}} |\psi(A)\rangle \right] \\ &= \sum_n \boxed{A^{[1]}} - \dots - \boxed{A^{[n-1]}} - \boxed{B^{[n]}} - \boxed{A^{[n+1]}} - \dots - \boxed{A^{[N]}} \end{aligned}$$



Matrix product states (MPS)

Ground state

$$|\psi(A)\rangle = \begin{array}{c} \boxed{A_L^{[1]}} \rightarrow \dots \rightarrow \boxed{A_L^{[n]}} \rightarrow \boxed{C^{[n]}} \leftarrow \boxed{A_R^{[n+1]}} \leftarrow \dots \leftarrow \boxed{A_R^{[N]}} \\ \uparrow \qquad \qquad \qquad \uparrow \qquad \qquad \qquad \uparrow \qquad \qquad \qquad \uparrow \end{array} \xRightarrow{\text{DMRG}} |\psi_0\rangle$$

$$\begin{array}{c} \boxed{A_L} \\ \updownarrow \\ \boxed{\bar{A}_L} \end{array} = \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \qquad \begin{array}{c} \boxed{A_R} \\ \updownarrow \\ \boxed{\bar{A}_R} \end{array} = \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right]$$

Matrix product states (MPS)

Ground state	$ \psi_0\rangle = \begin{array}{c} \boxed{A_L^{[1]}} \rightarrow \boxed{A_L^{[2]}} \rightarrow \dots \rightarrow \boxed{A_L^{[N]}} \rightarrow \boxed{C^{[N]}} \leftarrow \\ \uparrow \quad \uparrow \quad \uparrow \end{array}$
Quasiparticle Excitations [Van Damme et al. 2021]	Tangent space $\psi(B)\rangle = \sum_{n=1}^N \begin{array}{c} \boxed{A_L^{[1]}} \rightarrow \dots \rightarrow \boxed{A_L^{[n-1]}} \rightarrow \boxed{B^{[n]}} \rightarrow \boxed{A_L^{[n+1]}} \rightarrow \dots \rightarrow \boxed{A_L^{[N]}} \rightarrow \boxed{C^{[N]}} \leftarrow \\ \uparrow \quad \uparrow \quad \uparrow \end{array}$ Gauge freedom $B^{[n]} \rightarrow B^{[n]} + A_L^{[n]} Y^{[n]} - Y^{[n-1]} A_L^{[n]}$ Parametrization $B(X)$ such that (i) $\langle \psi_0 \psi(X) \rangle = 0$, (ii) $X \mapsto \psi(X)\rangle$ injective $\begin{array}{c} \boxed{B^{[n]}} \\ \downarrow \end{array} = \begin{array}{c} \rightarrow \boxed{V_L^{[n]}} \rightarrow \boxed{X^{[n]}} \leftarrow \\ \uparrow \end{array} \quad \text{with} \quad \begin{array}{c} \rightarrow \boxed{V_L^{[n]}} \rightarrow \\ \updownarrow \\ \leftarrow \boxed{\bar{A}_L^{[n]}} \leftarrow \end{array} = 0$ $ \psi(X)\rangle = \sum_{n=1}^N \begin{array}{c} \boxed{A_L^{[1]}} \rightarrow \dots \rightarrow \boxed{A_L^{[n-1]}} \rightarrow \boxed{V_L^{[n]}} \rightarrow \boxed{X^{[n]}} \rightarrow \boxed{A_R^{[n+1]}} \leftarrow \dots \leftarrow \boxed{A_R^{[N]}} \\ \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \end{array}$ $\Rightarrow \langle \psi_0 \psi(X^{[n]}) \rangle = 0$ $\Rightarrow \langle \psi(\bar{Y}^{[m]}) \psi(X^{[n]}) \rangle = \delta_{m,n} (\bar{Y}^{[n]} X^{[n]})$

Restoring momentum with PBC Hamiltonian

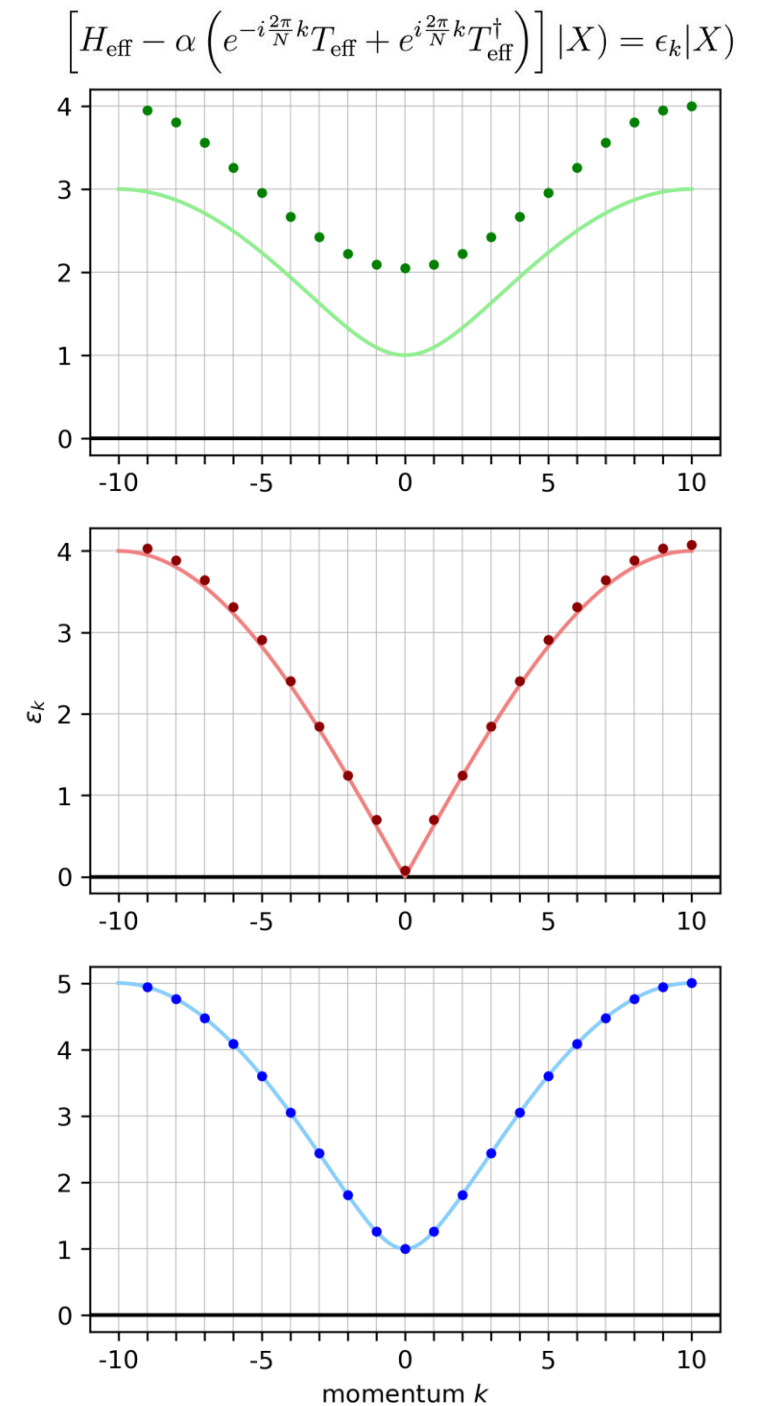
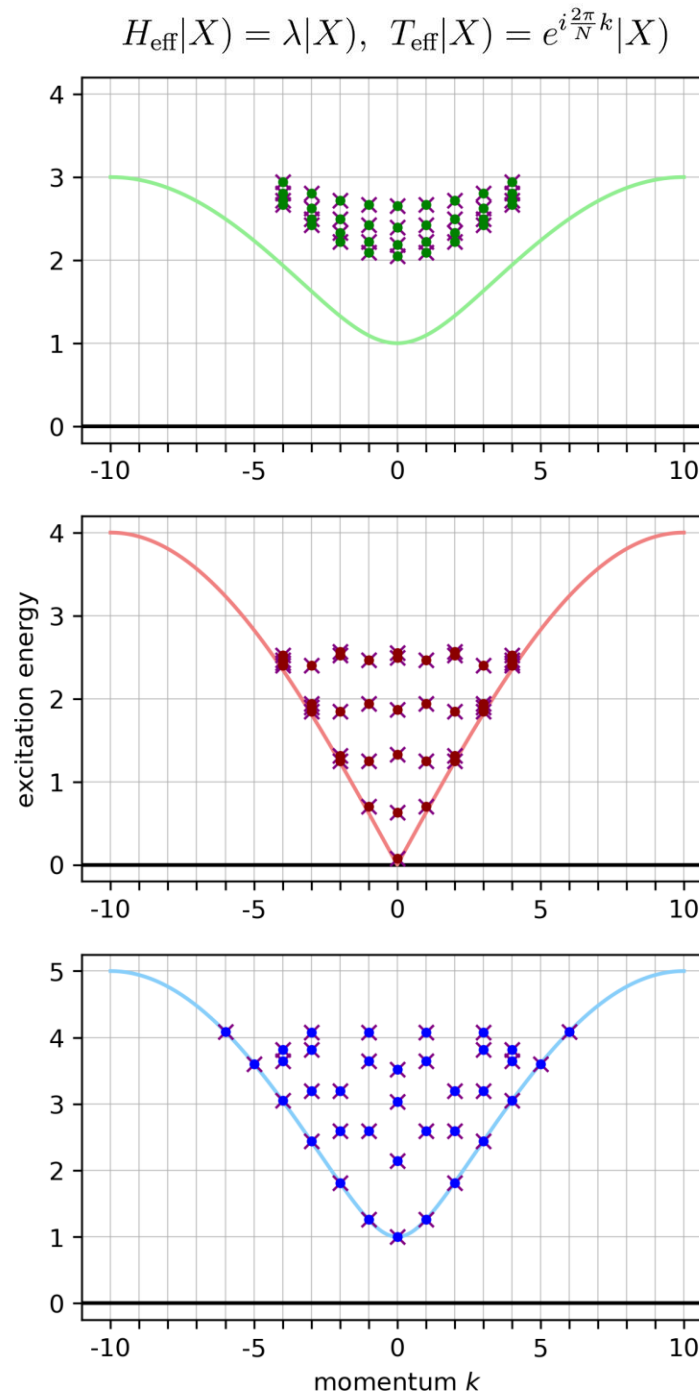
(a) DMRG + VQPE ($D_{\max} = 64$)

- $g = 0.5$
- $g = 1.0$
- $g = 1.5$

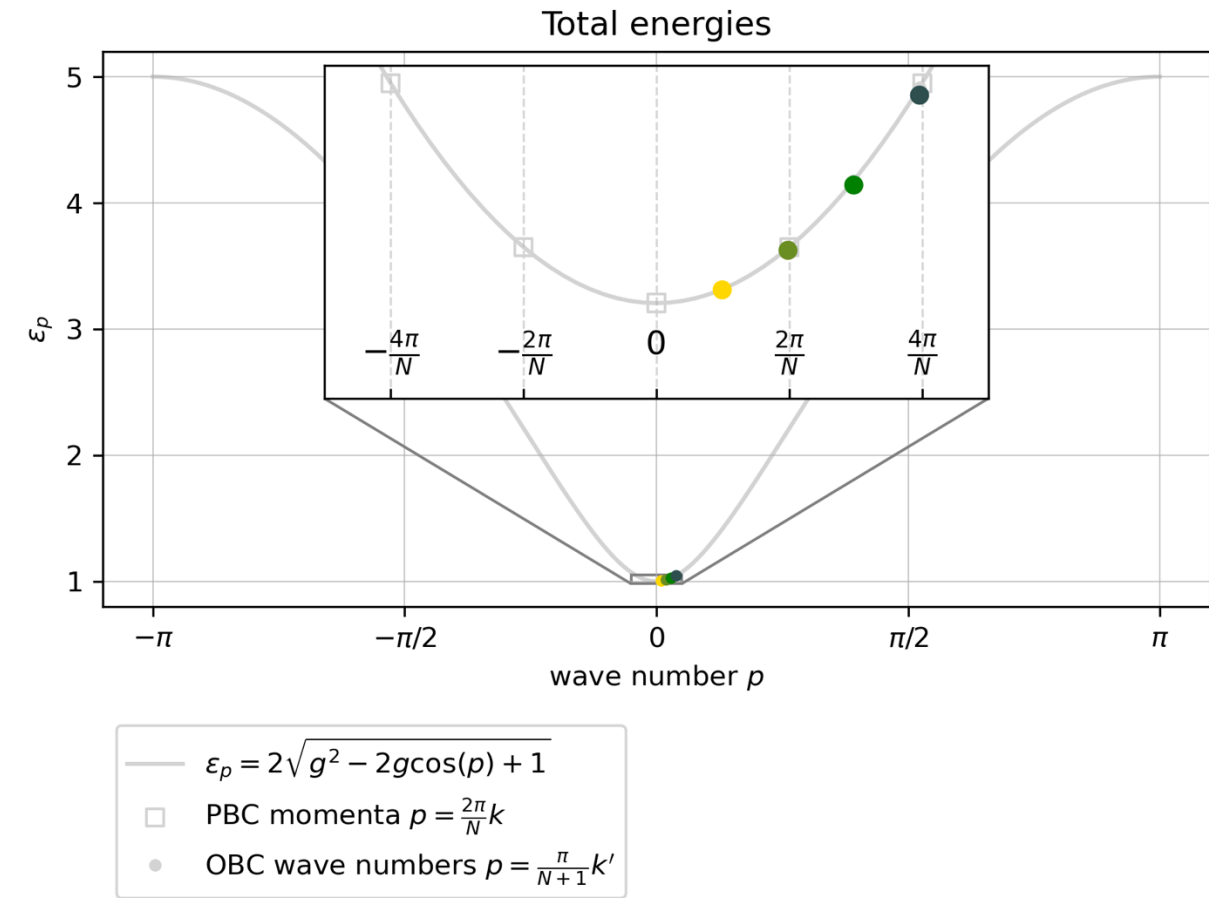
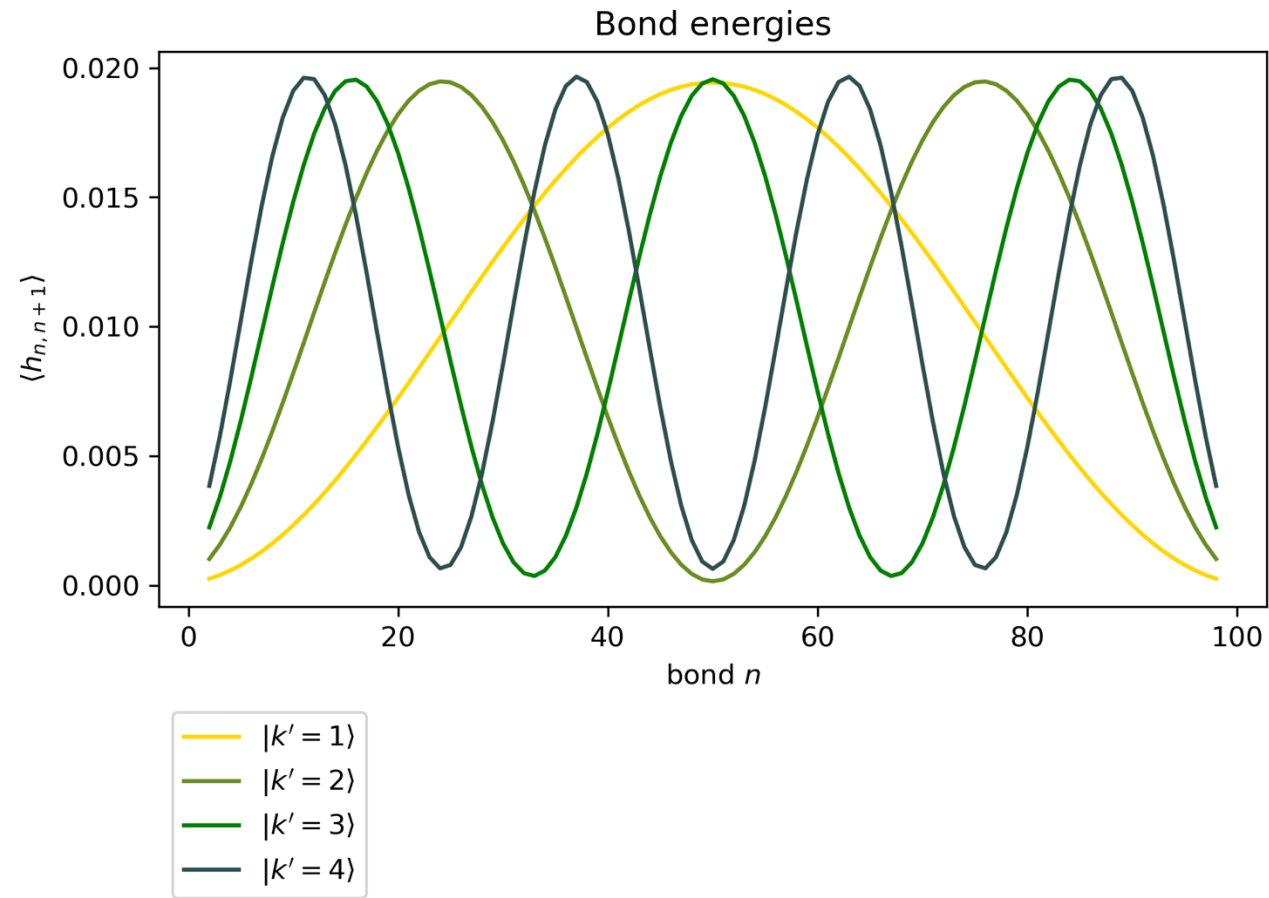
✕ Exact diagonalization

(a)(b) $\epsilon_k = 2\sqrt{g^2 - 2g\cos\left(\frac{2\pi}{N}k\right) + 1}$

- $g = 0.5$
- $g = 1.0$
- $g = 1.5$



Standing waves with OBC Hamiltonian

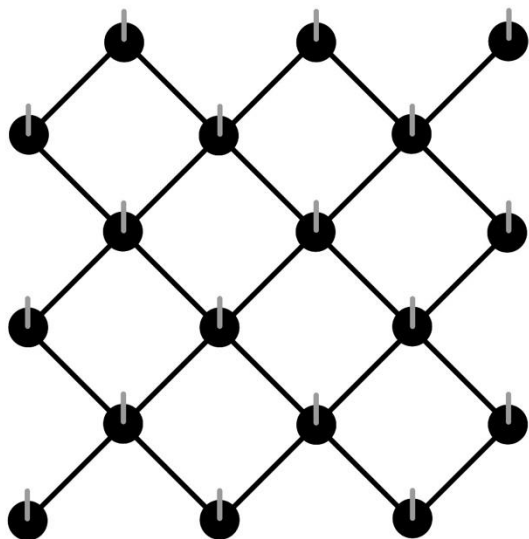


$$|k\rangle = \frac{1}{\sqrt{N}} \sum_n e^{i\frac{2\pi k}{N}n} |n\rangle$$

$$|k'\rangle = \sqrt{\frac{2}{N+1}} \sum_n \sin\left(\frac{\pi k'}{N+1}n\right) |n\rangle$$

Projected entangled pair states (PEPS)

$$|\psi(A)\rangle =$$

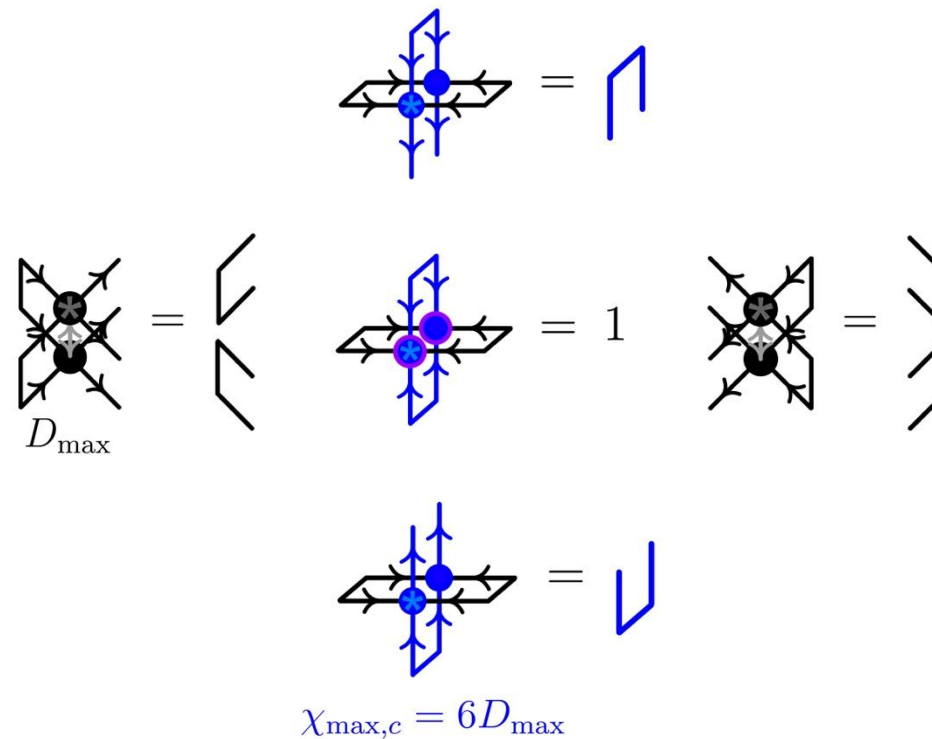
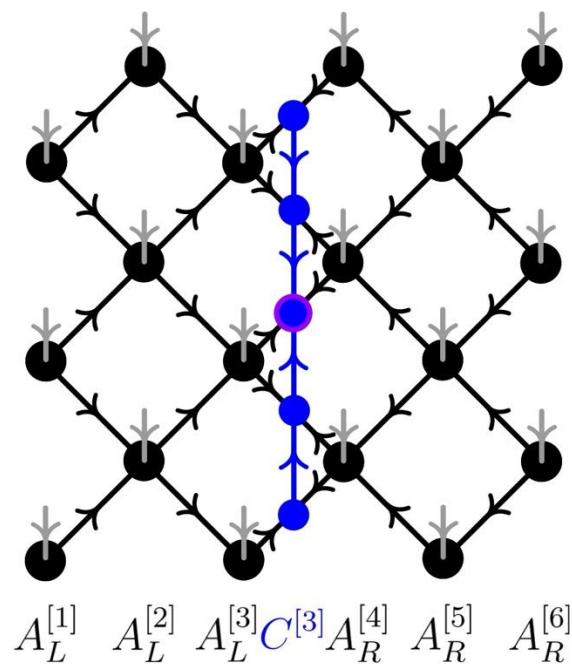


isometric PEPS

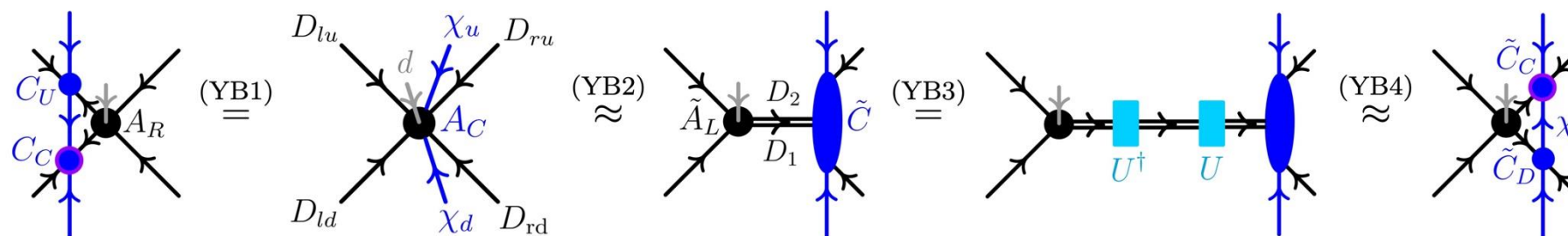
Conventional square lattice [Zaletel and Pollmann 2020]

Diagonal square lattice [Sappler et al. 2025]

$$|\psi(A_L, C, A_R)\rangle =$$

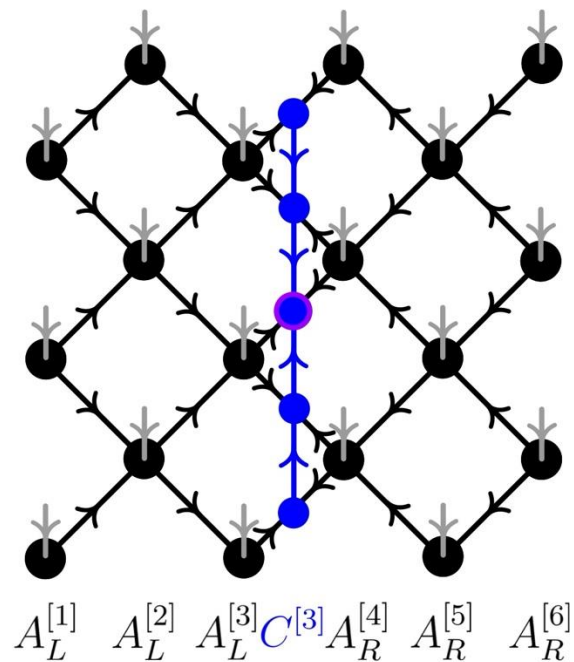


Yang-Baxter move

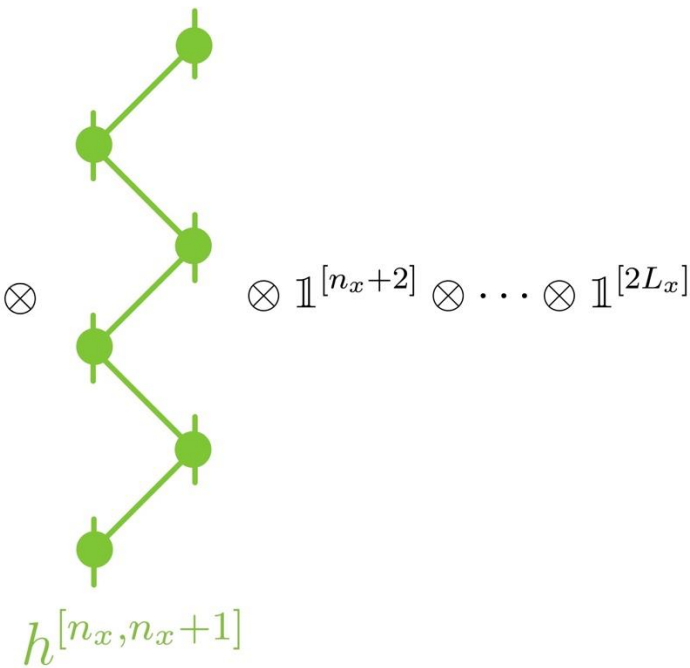


Energy expectation value

$$|\psi(A_L, C, A_R)\rangle =$$

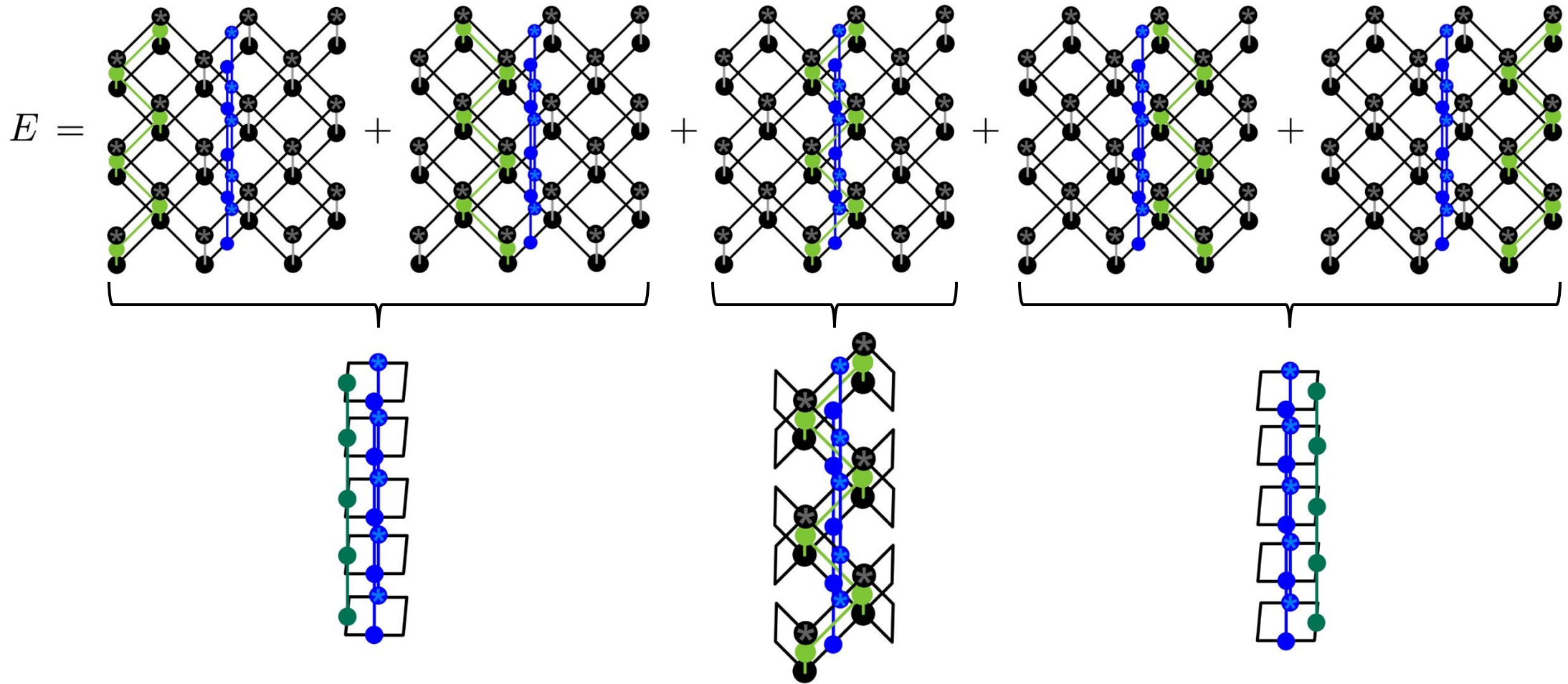


$$H = \sum_{n_x=1}^{2L_x-1} \mathbb{1}^{[1]} \otimes \dots \otimes \mathbb{1}^{[n_x-1]} \otimes$$



$$\otimes \mathbb{1}^{[n_x+2]} \otimes \dots \otimes \mathbb{1}^{[2L_x]}$$

Energy expectation value



DMRG² [Lin et al. 2022]

right sweep

for x in range(L_x):

eigensolver for $H_{\text{eff},0}$

up sweep

for n_y in range(N_y):

theta2 = Lanczos($H_{\text{eff},2}$)

SVD(theta2)

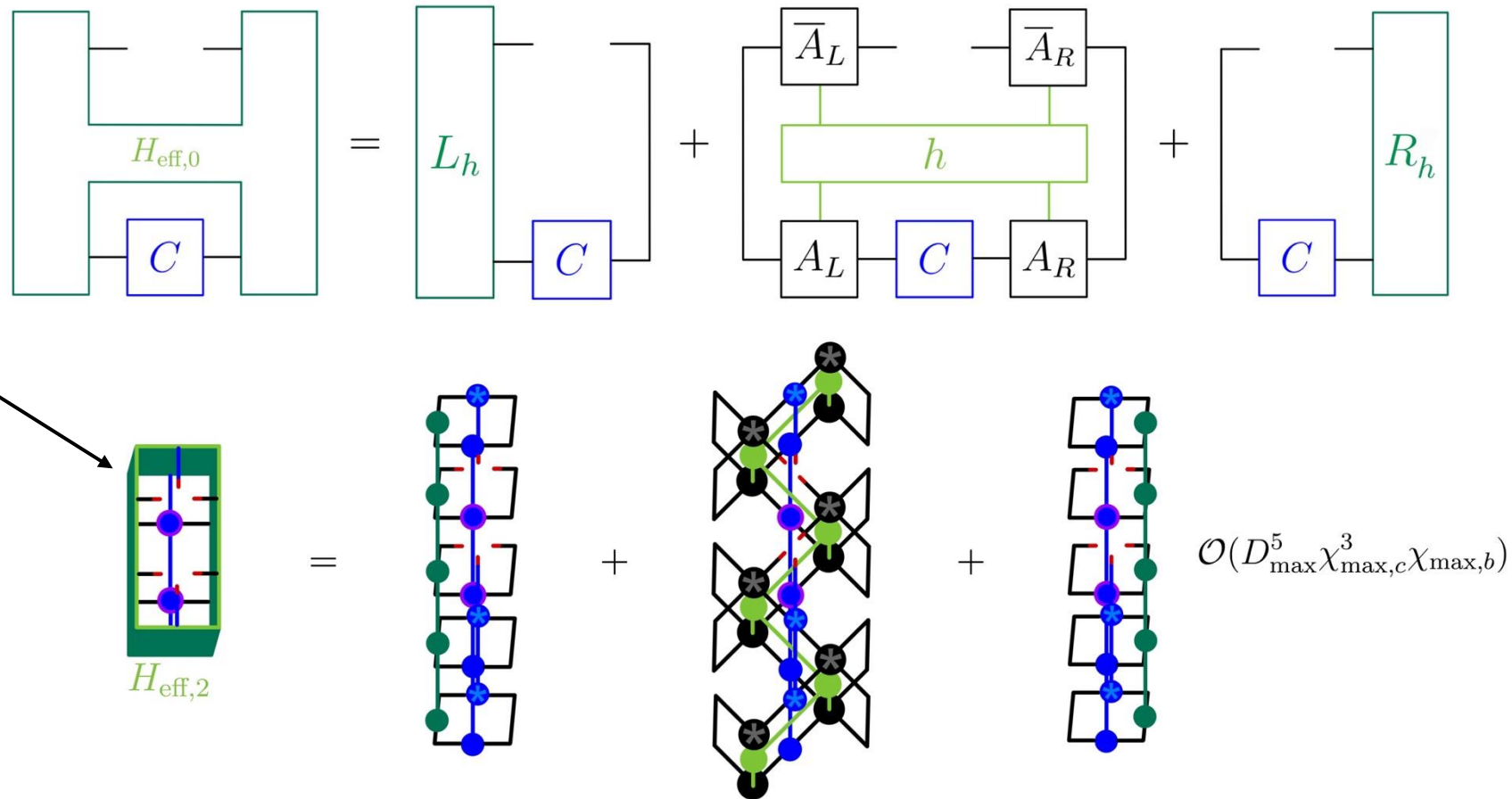
down_environment_update()

down sweep ...

yang_baxter_move()

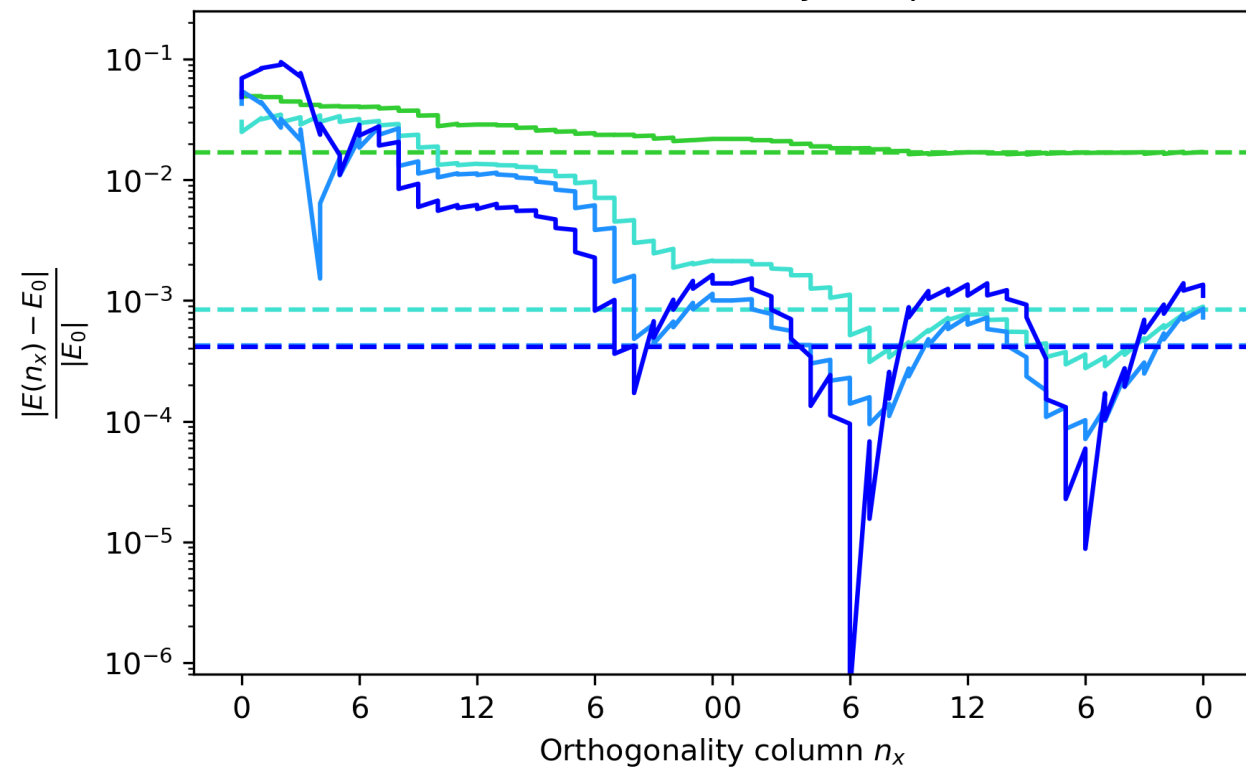
left_boundary_compression()

left sweep ...

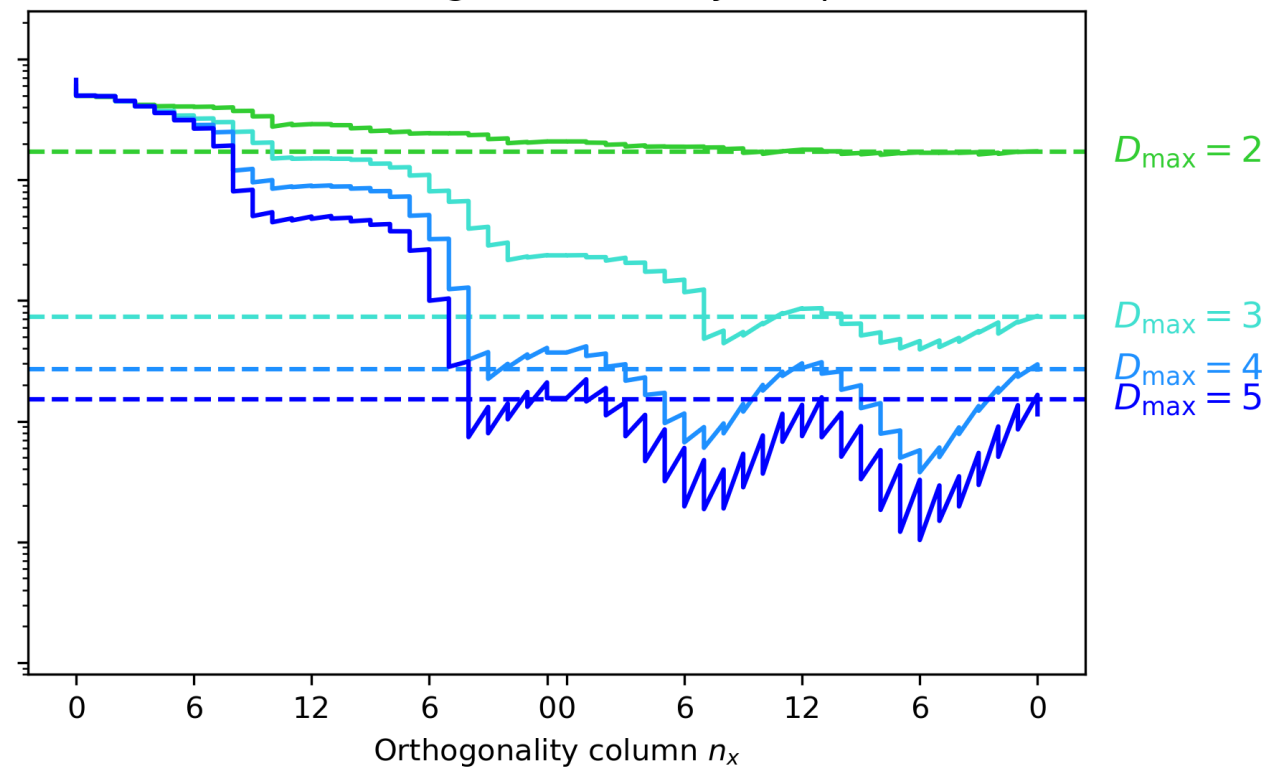


$$\text{DMRG}^2 \quad \chi_{\max,c} = 6D_{\max}; \quad \chi_{\max,b} = 6D_{\max}^2$$

(a) Variational boundary compression



(b) New bulk-weighted boundary compression

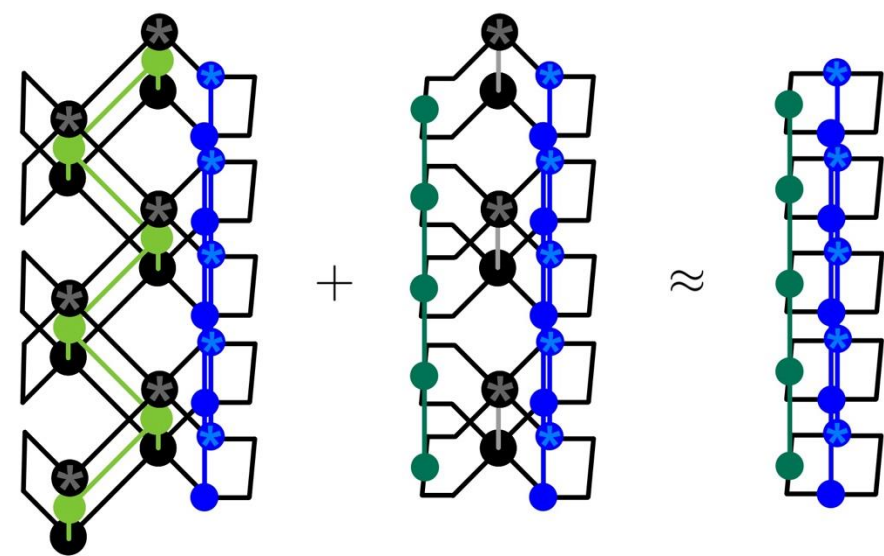


boundary MPS

$$\left(L_h^{[n_x]} \right| = \left(\begin{array}{c} \overline{A}_L^{[n_x-1]} \overline{A}_L^{[n_x]} \\ h^{[n_x-1, n_x]} \\ A_L^{[n_x-1]} A_L^{[n_x]} \end{array} \right| + \left(L_h^{[n_x-1]} \right| \begin{array}{c} \overline{A}_L^{[n_x]} \\ A_L^{[n_x]} \end{array}$$

$$\chi_{\max, b}^{\text{exact}} = D_{\max}^{2(L_y-1)} \rightarrow \chi_{\max, b} = 6D_{\max}^2$$

New bulk-weighted boundary compression

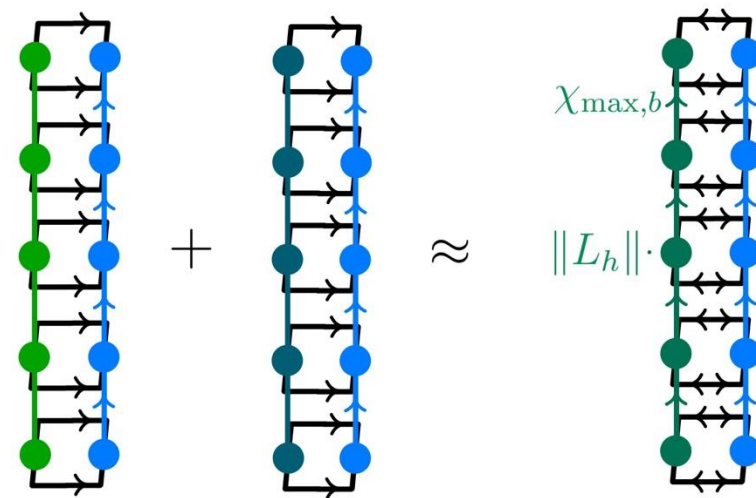


New bulk-weighted boundary compression

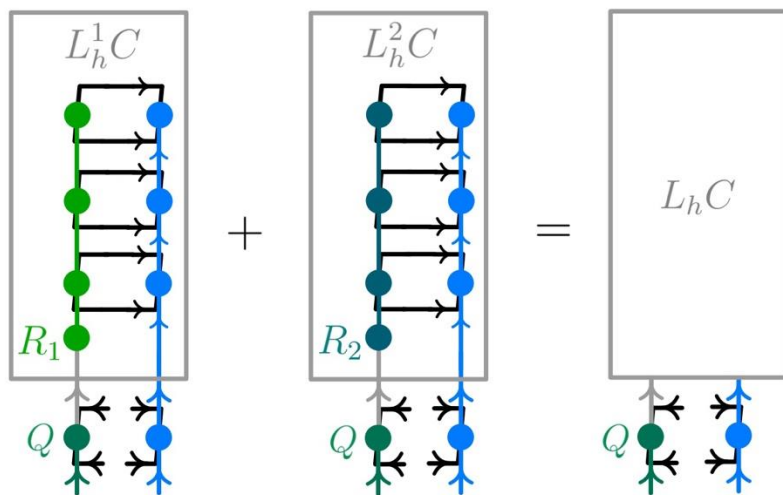
1) L_h^1 L_h^2 $\stackrel{\text{stacked QR}}{=} \begin{matrix} R_1 & R_2 \\ Q & Q \end{matrix}$

$$(L_h^1 \ L_h^2) = QR = Q(R_1 \ R_2)$$

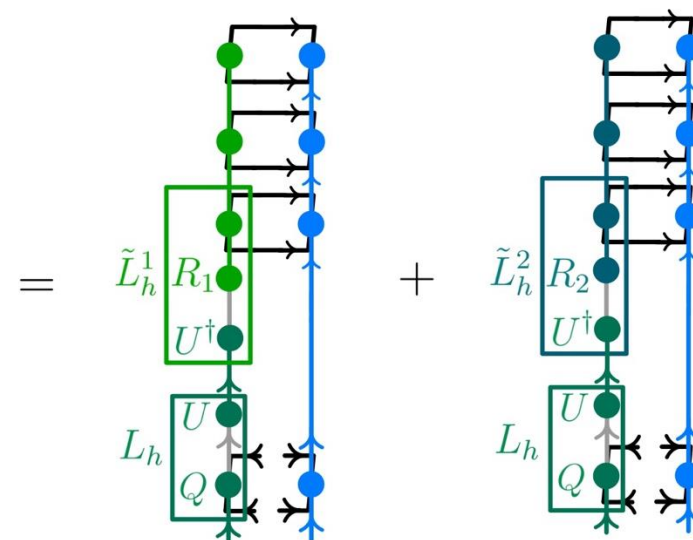
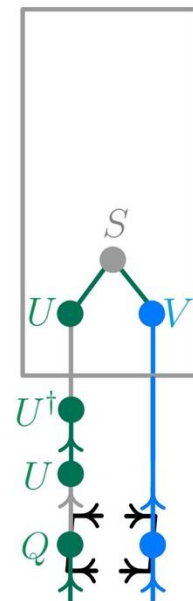
$$\mathcal{O}(D_{\max}^6 \chi_{\max,b}^3)$$



2)

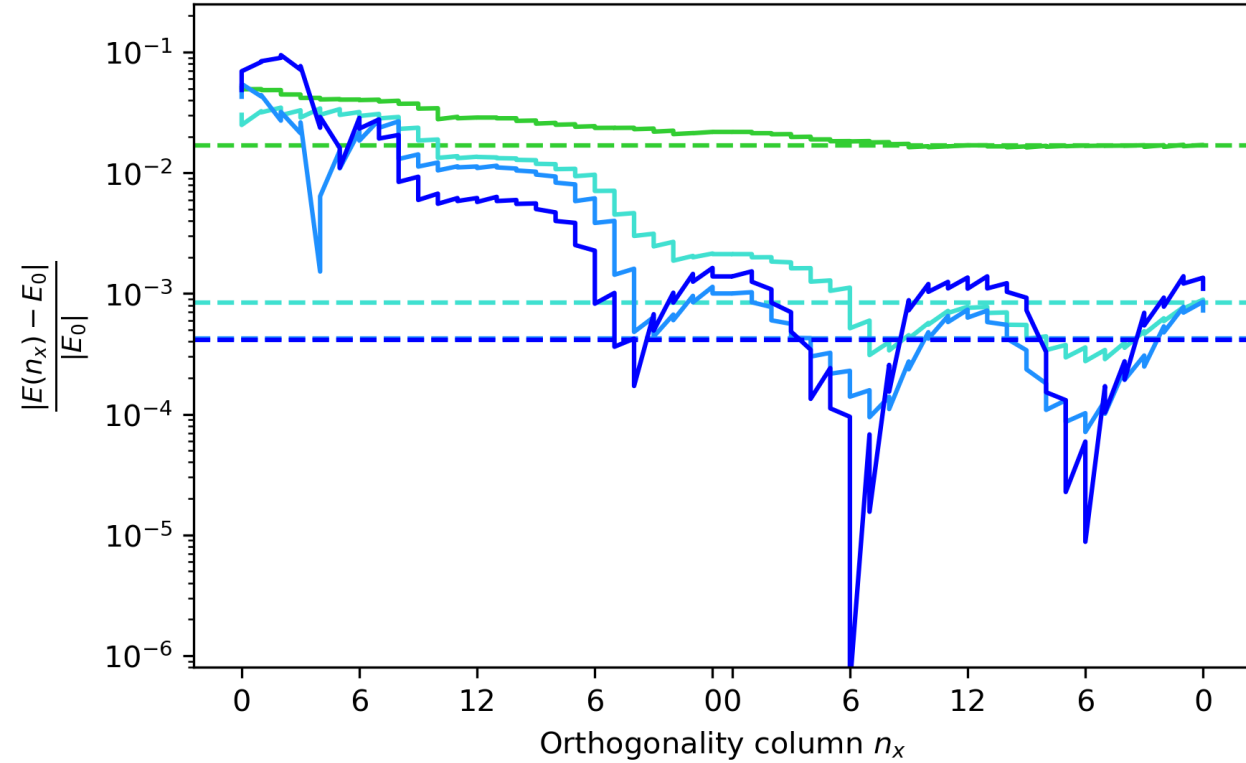


$$\stackrel{\text{tSVD}}{\approx} \chi_{\max,b}$$

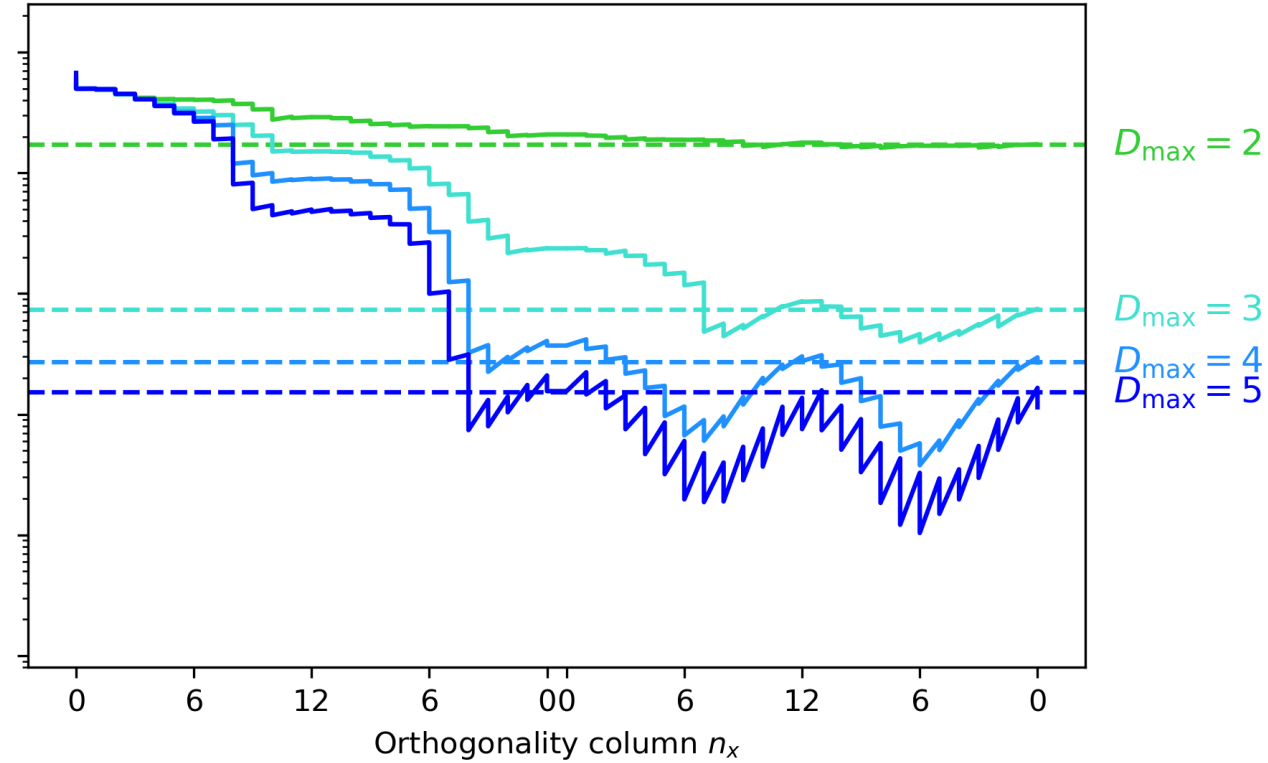


$$\text{DMRG}^2 \quad \chi_{\max,c} = 6D_{\max}; \quad \chi_{\max,b} = 6D_{\max}^2$$

(a) Variational boundary compression



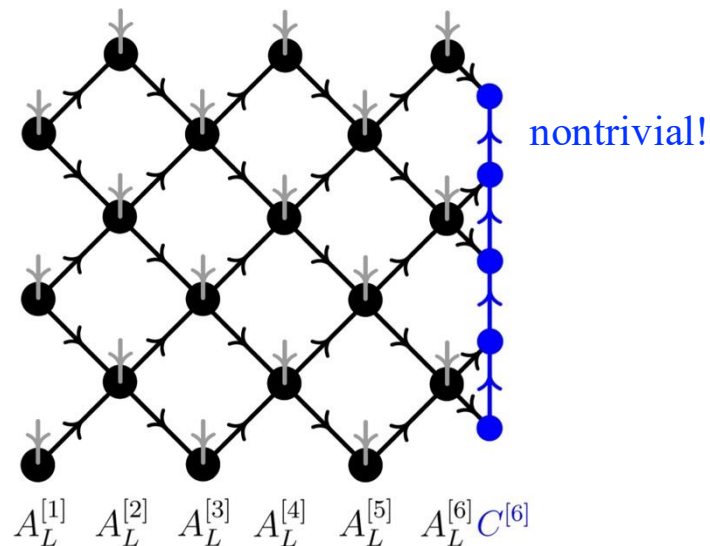
(b) New bulk-weighted boundary compression



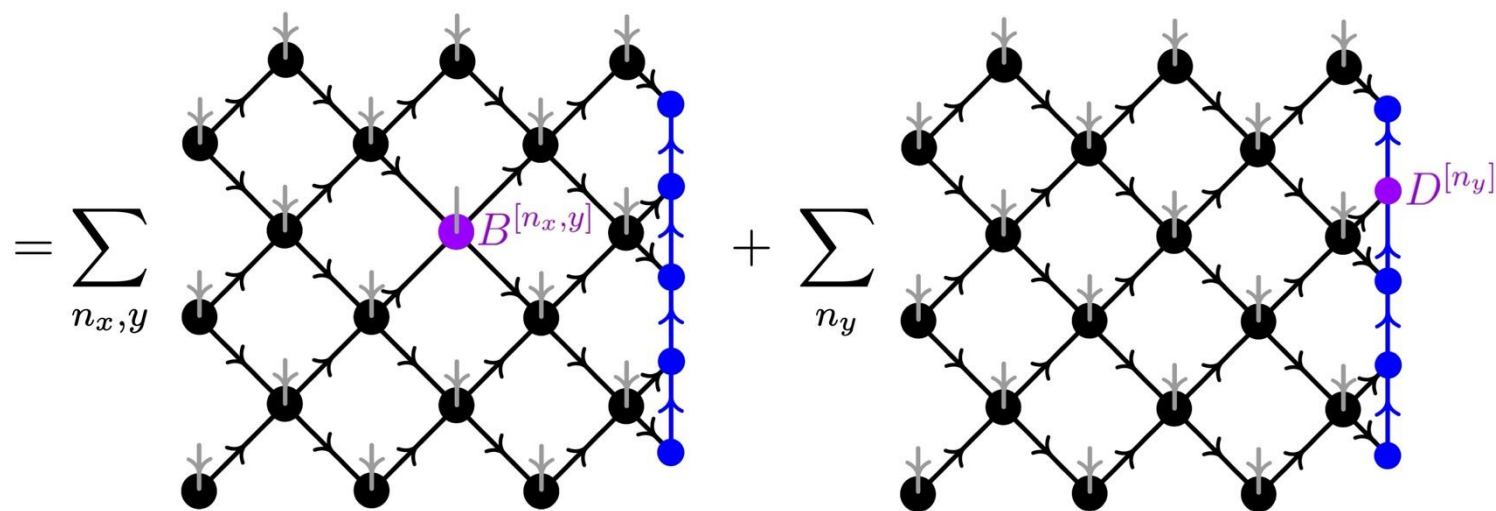
$\Rightarrow$ Bulk-weighted boundary compression better than variational one for $D_{\max} \geq 4$!

isoPEPS tangent space

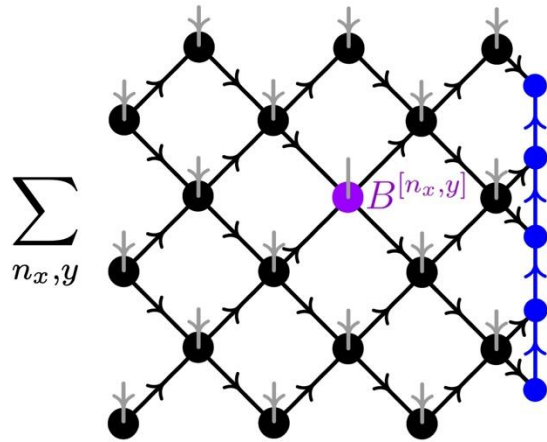
$$|\psi(A_L, C)\rangle =$$



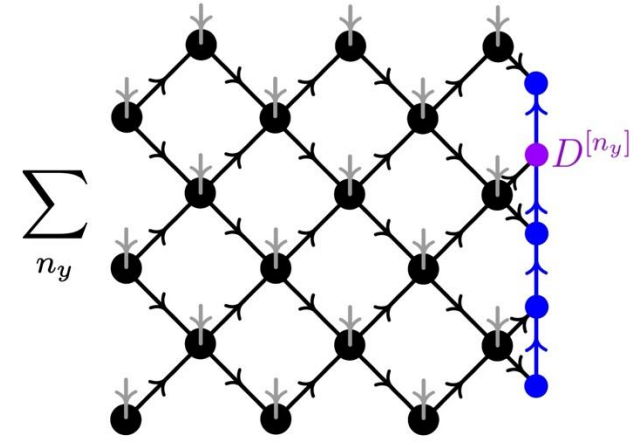
$$|\psi(B, D; A_L, C)\rangle = \sum_{n_x, y} \sum_{s, \alpha, \beta, \gamma, \delta} B_{\alpha\beta\gamma\delta}^{[n_x, y]s} \left[\partial_{(A_L^{[n_x, y]})_{\alpha\beta\gamma\delta}^s} |\psi(A_L, C)\rangle \right] + \sum_{n_y} \sum_{d, l, r, u} D_{dlru}^{[n_y]} \left[\partial_{C_{dlru}^{[2L_x, n_y]}} |\psi(A_L, C)\rangle \right]$$



Tangent space parametrization $B(X), D(X)$ such that (i) $\langle \psi_0 | \psi(X) \rangle = 0$, (ii) $X \mapsto |\psi(X)\rangle$ injective



+



$$B^{[n_x, y]} = \text{Diagram} \quad \text{with} \quad \text{Diagram} = 0$$

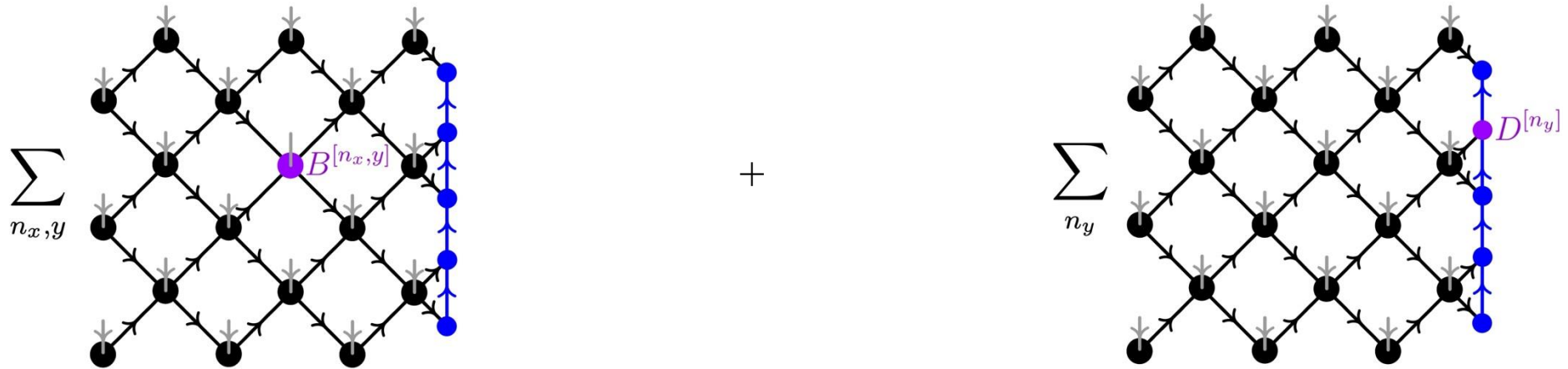
The first diagram shows a node with four incoming arrows (two black, two grey) and a purple arrow pointing to it, labeled $V_L^{[n_x, y]}$ and $X^{[n_x, y]}$. The second diagram shows a node with four incoming arrows (two black, two grey) and a purple arrow pointing away from it, labeled $\bar{A}_L^{[n_x, y]}$ and $V_L^{[n_x, y]}$.

$$\Rightarrow \langle \psi_0 | \psi(X^{[n_x, y]}) \rangle = 0$$

$$\Rightarrow \langle \psi(\bar{X}^{[n_y]}) | \psi(X^{[n_x, y]}) \rangle = 0$$

$$\Rightarrow \langle \psi(\bar{X}^{[\tilde{n}_x, \tilde{y}]}) | \psi(X^{[n_x, y]}) \rangle \propto \delta_{\tilde{n}_x, n_x} \delta_{\tilde{y}, y}$$

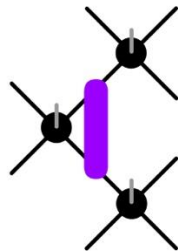
Tangent space parametrization $B(X), D(X)$ such that (i) $\langle \psi_0 | \psi(X) \rangle = 0$, (ii) $X \mapsto |\psi(X)\rangle$ injective



$$B^{[n_x, y]} = \text{diagram of two nodes connected by a purple line} \quad \text{with} \quad \text{diagram of a node with four incoming arrows} = 0$$

$V_L^{[n_x, y]} X^{[n_x, y]}$ $\overline{A}_L^{[n_x, y]}$ $V_L^{[n_x, y]}$

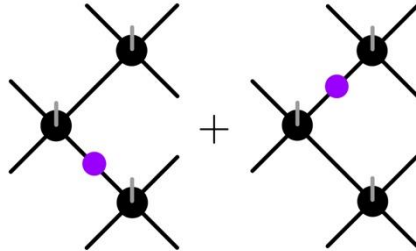
projects out



$$A_L Y_{du}$$

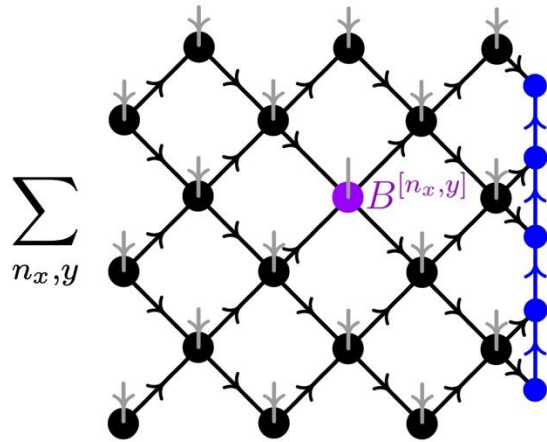
generic zero modes

vs.

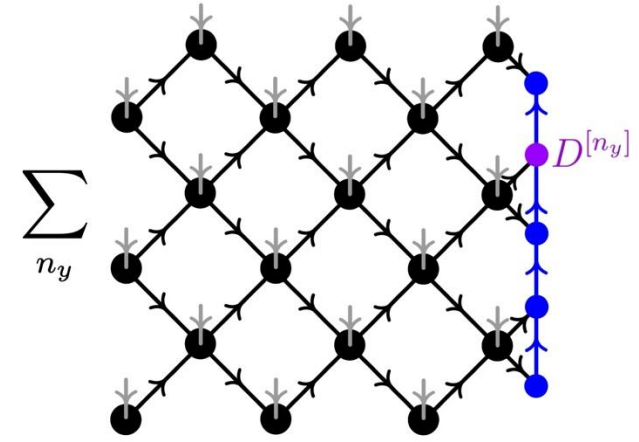


$$A_L(Y_d \otimes \mathbb{1}_u) + A_L(\mathbb{1}_d \otimes Y_u)$$

Tangent space parametrization $B(X), D(X)$ such that (i) $\langle \psi_0 | \psi(X) \rangle = 0$, (ii) $X \mapsto |\psi(X)\rangle$ injective



+



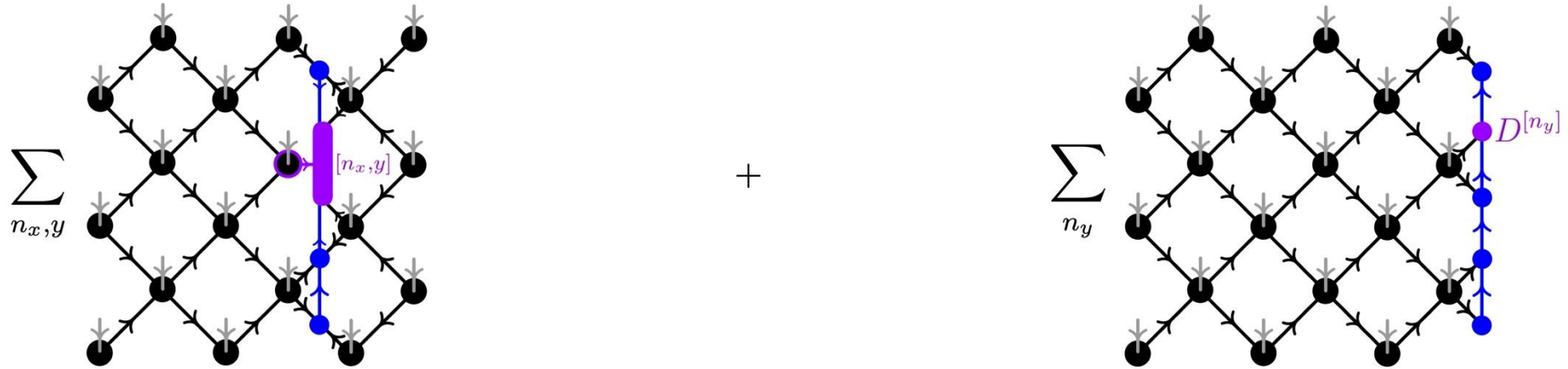
$$B^{[n_x, y]} = \begin{array}{c} \text{Diagram of a node with four incoming arrows and a purple line} \\ V_L^{[n_x, y]} \quad X^{[n_x, y]} \end{array} \quad \text{with} \quad \begin{array}{c} \text{Diagram of a node with four incoming arrows and a purple line} \\ \bar{A}_L^{[n_x, y]} \quad V_L^{[n_x, y]} \end{array} = 0$$

$$\Rightarrow \langle \psi_0 | \psi(X^{[n_x, y]}) \rangle = 0$$

$$\Rightarrow \langle \psi(\bar{X}^{[n_y]}) | \psi(X^{[n_x, y]}) \rangle = 0$$

$$\Rightarrow \langle \psi(\bar{X}^{[\tilde{n}_x, \tilde{y}]}) | \psi(X^{[n_x, y]}) \rangle \propto \delta_{\tilde{n}_x, n_x} \delta_{\tilde{y}, y}$$

Tangent space parametrization $B(X), D(X)$ such that (i) $\langle \psi_0 | \psi(X) \rangle = 0$, (ii) $X \mapsto |\psi(X)\rangle$ injective



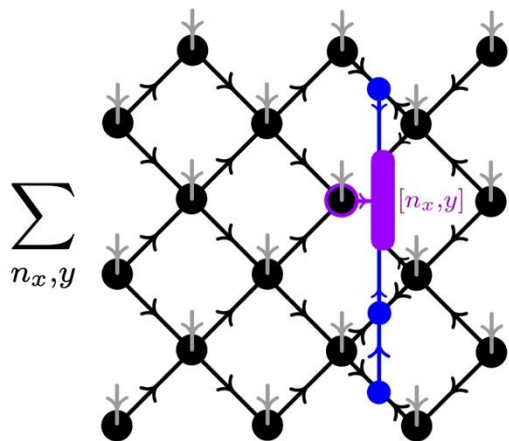
$$B^{[n_x, y]} = V_L^{[n_x, y]} \text{ (diagram of a node with four incoming arrows) } X^{[n_x, y]} \text{ (diagram of a vertical bar) with } \overline{A}_L^{[n_x, y]} \text{ (diagram of a node with four outgoing arrows) } V_L^{[n_x, y]} = 0$$

$$\Rightarrow \langle \psi_0 | \psi(X^{[n_x, y]}) \rangle = 0$$

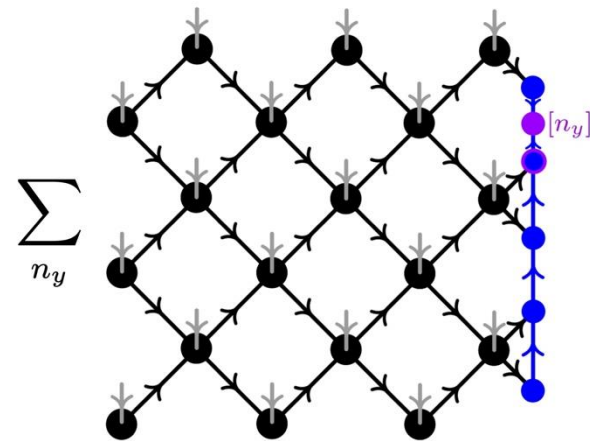
$$\Rightarrow \langle \psi(\overline{X}^{[n_y]}) | \psi(X^{[n_x, y]}) \rangle = 0$$

$$\Rightarrow \langle \psi(\overline{X}^{[\tilde{n}_x, \tilde{y}]}) | \psi(X^{[n_x, y]}) \rangle = \delta_{\tilde{n}_x, n_x} \delta_{\tilde{y}, y} (\overline{X}^{[n_x, y]} | X^{[n_x, y]})$$

Quasiparticle excitation ansatz



+



$$B^{[n_x, y]} = V_L^{[n_x, y]} \text{ with } \overline{A}_L^{[n_x, y]} = 0$$

A diagram showing a node with four arrows (two incoming, two outgoing) and a vertical chain of blue and purple nodes. A purple node is labeled $X^{[n_x, y]}$. The equation is $B^{[n_x, y]} = V_L^{[n_x, y]}$ with $\overline{A}_L^{[n_x, y]} = 0$.

$$\Rightarrow \langle \psi_0 | \psi(X^{[n_x, y]}) \rangle = 0$$

$$\Rightarrow \langle \psi(\overline{X}^{[n_y]}) | \psi(X^{[n_x, y]}) \rangle = 0$$

$$\Rightarrow \langle \psi(\overline{X}^{[\tilde{n}_x, \tilde{y}]}) | \psi(X^{[n_x, y]}) \rangle = \delta_{\tilde{n}_x, n_x} \delta_{\tilde{y}, y} (\overline{X}^{[n_x, y]} | X^{[n_x, y]})$$

$$D^{[n_y]} = \text{with } \overline{C}_D^{[n_y]} = 0$$

A diagram showing a node with four arrows (two incoming, two outgoing) and a vertical chain of blue and purple nodes. A purple node is labeled $X^{[n_y]}$. The equation is $D^{[n_y]} =$ with $\overline{C}_D^{[n_y]} = 0$.

$$\Rightarrow \langle \psi_0 | \psi(X^{[n_y]}) \rangle = 0$$

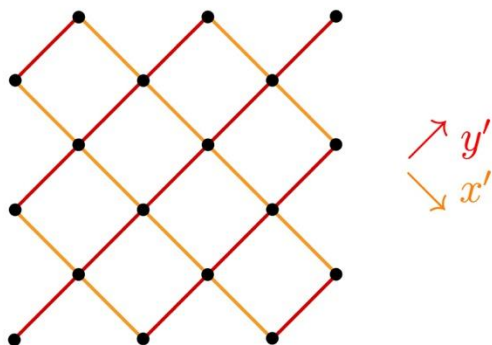
$$\Rightarrow \langle \psi(\overline{X}^{[\tilde{n}_y]}) | \psi(X^{[n_y]}) \rangle = \delta_{\tilde{n}_y, n_y} (\overline{X}^{[n_y]} | X^{[n_y]})$$

Which physics do we expect?

Paramagnetic phase $g = 3.5$

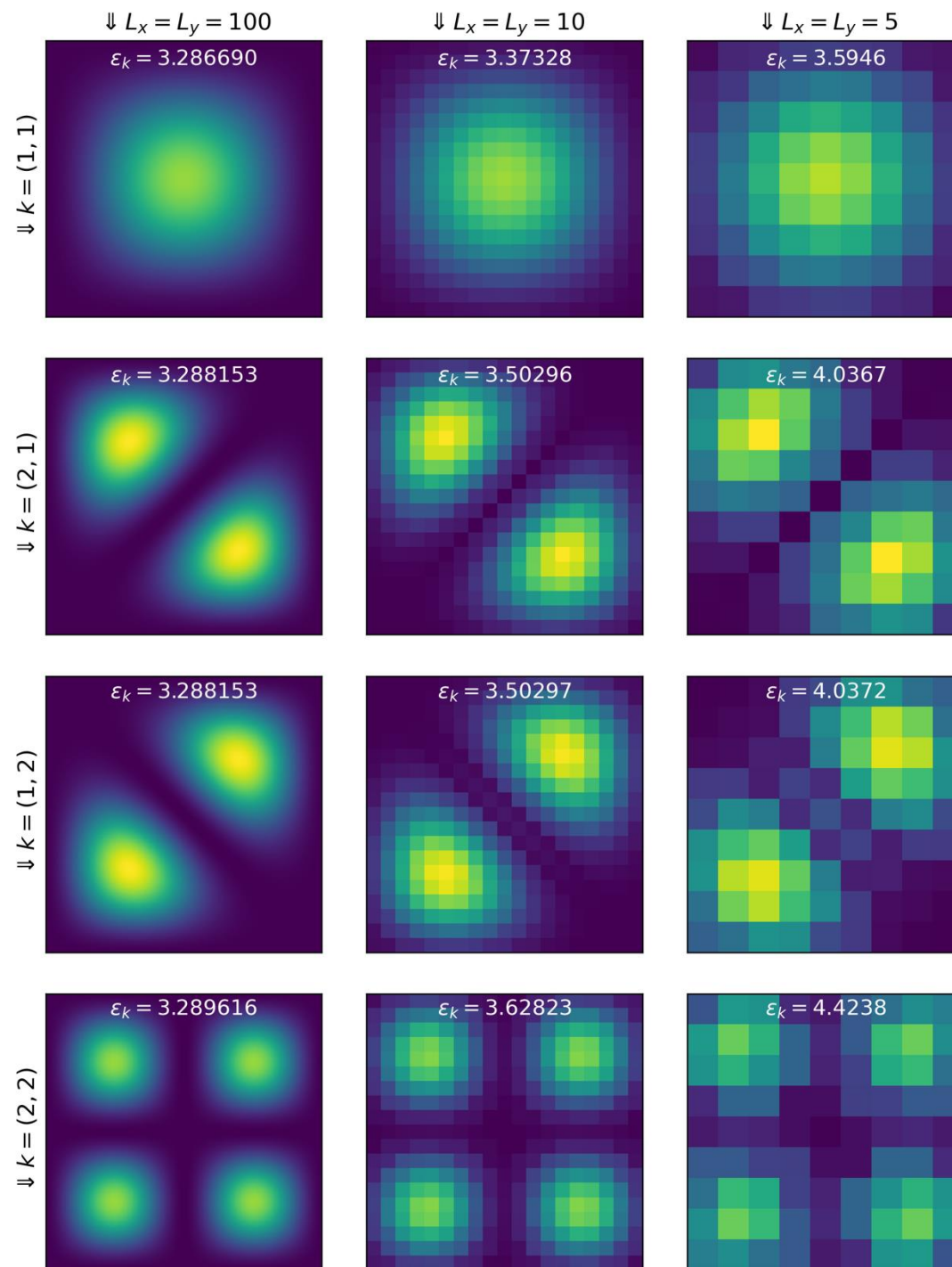
$$H_{\text{eff}} = \mu \sum_n |n\rangle \langle n| - t \sum_{\langle n, m \rangle} (|n\rangle \langle m| + |m\rangle \langle n|)$$

$$\mu_n = 2g + \frac{J^2}{4g} z_n, \quad t = 1$$



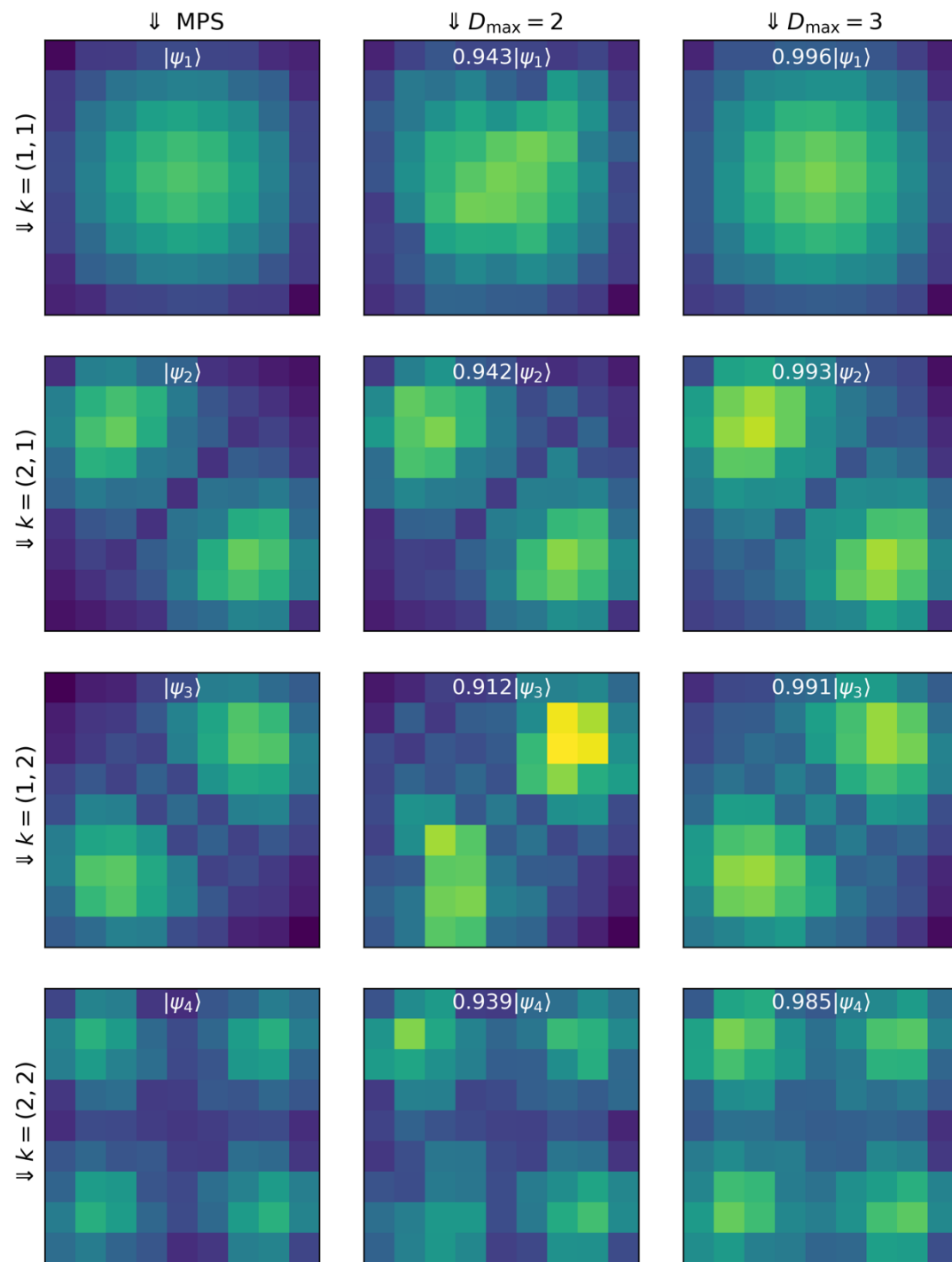
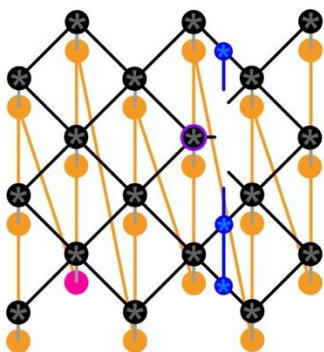
bond energies $\langle h_{n,m} \rangle$

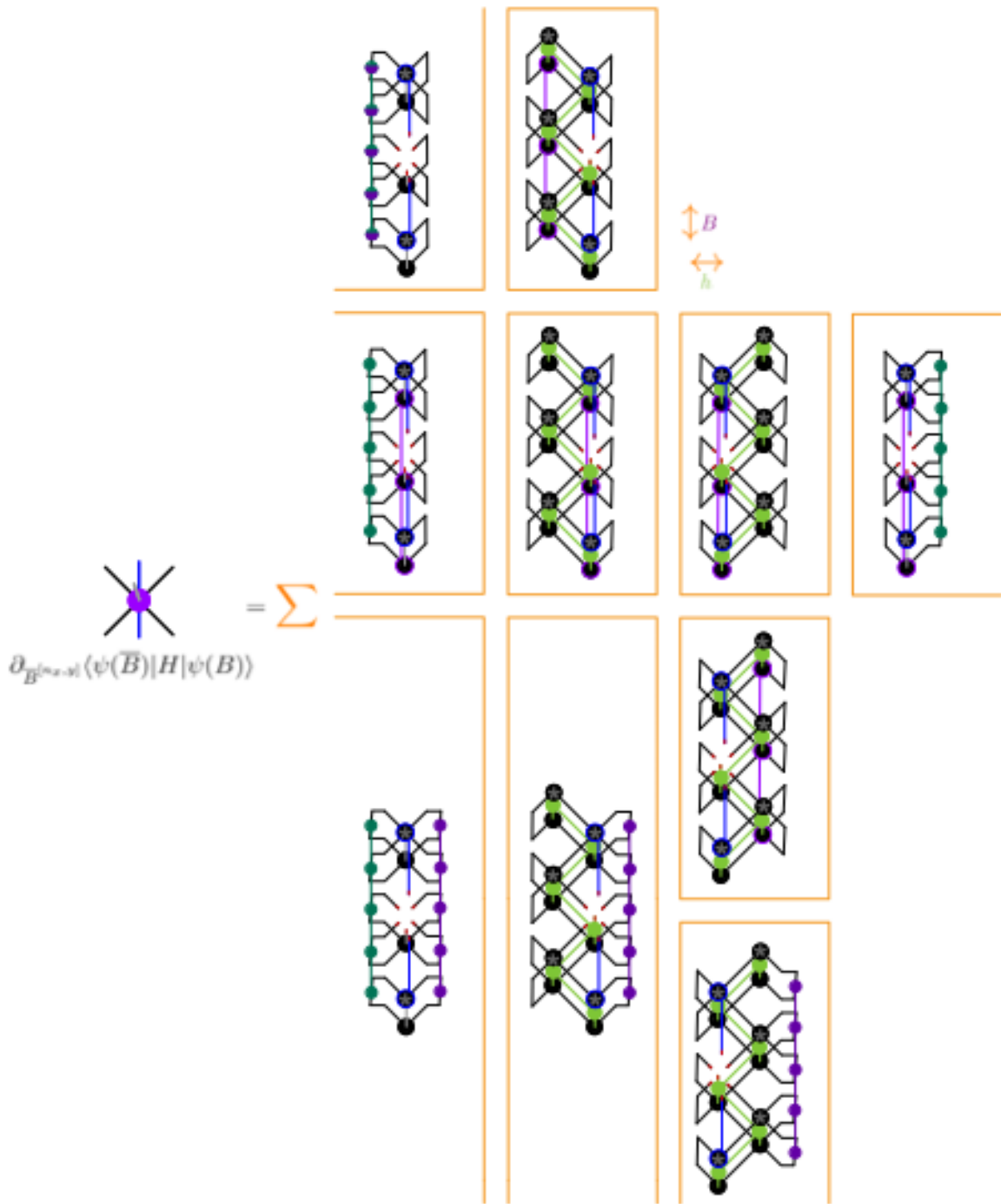
with $h_{n,m} = \frac{\mu}{2} |n\rangle \langle n| + \frac{\mu}{2} |m\rangle \langle m| - t (|n\rangle \langle m| + |m\rangle \langle n|)$



Can the ansatz represent the excitations?

$$X_{\text{MPS}}^{[n_x, y]} = \sum_{n=1}^{2L_x L_y} \partial_{\bar{X}^{[n_x, y]}} \langle \text{isoPEPS}(\bar{X}) | \text{MPS}(B^{[n]}) \rangle = \sum_{n=1}^{2L_x L_y}$$

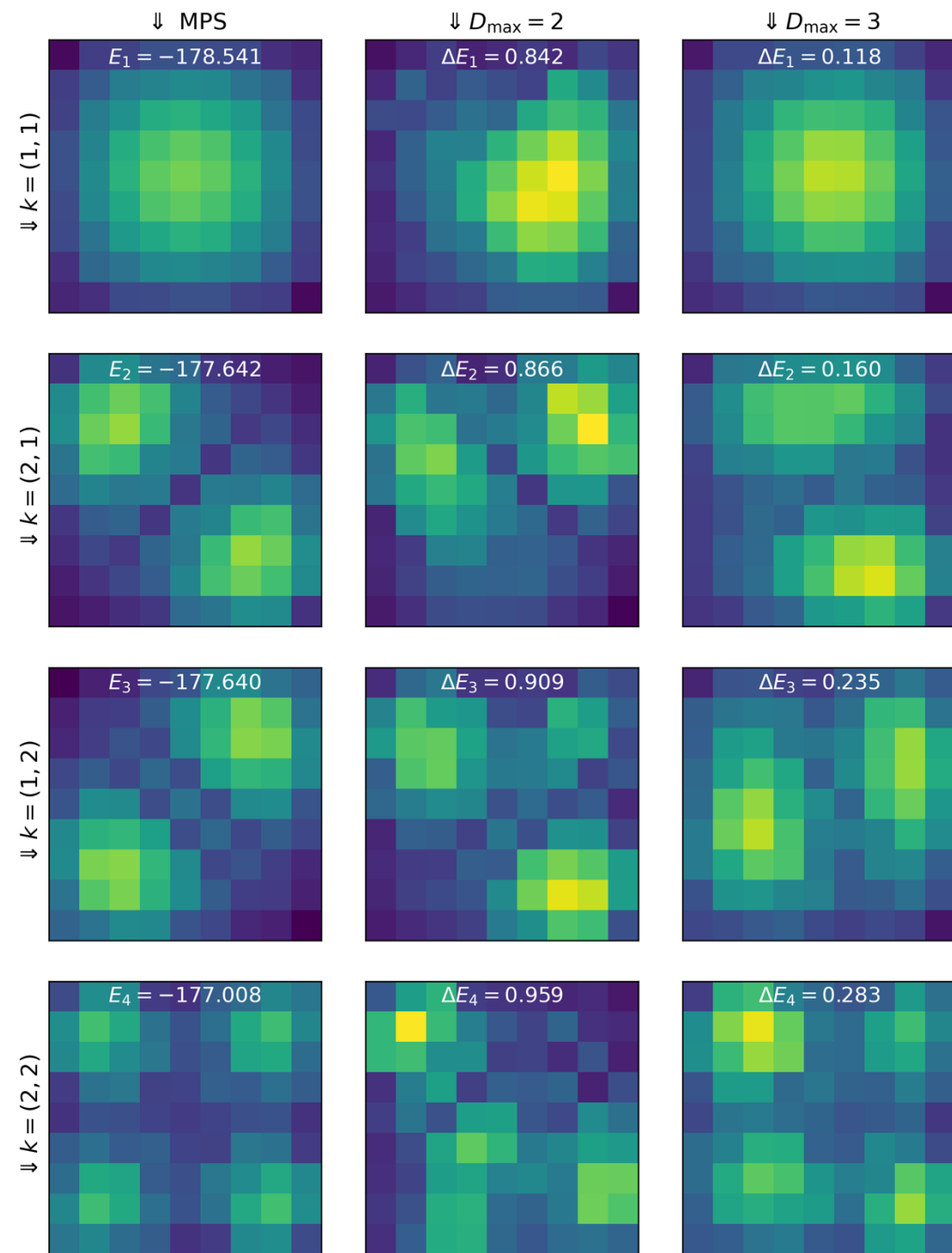
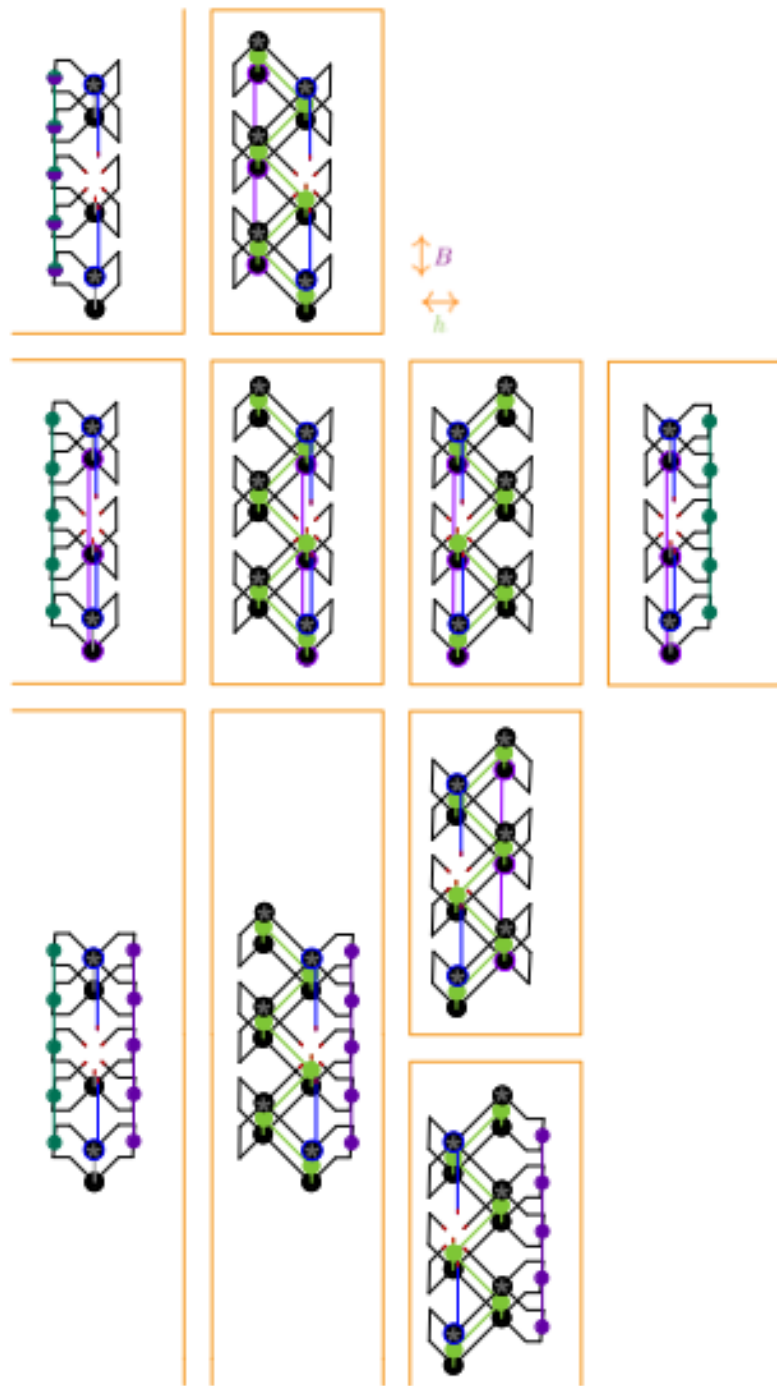




Can we find the representations from effective Hamiltonian?

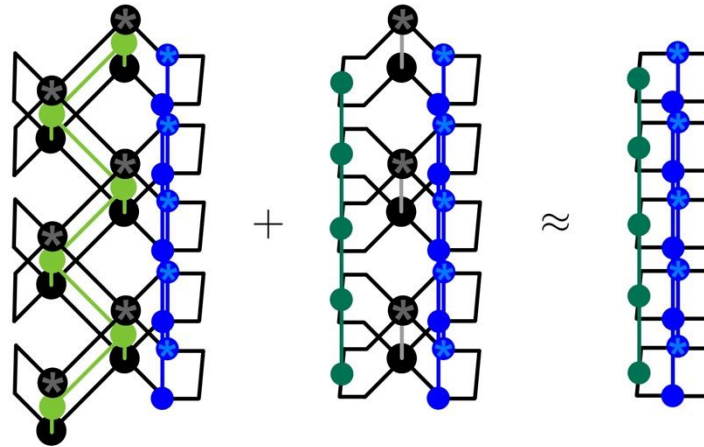
$$\begin{aligned} \left(L_{Bh}^{[n_x]} \right) &= \begin{pmatrix} \overline{A}_L^{[n_x-1]} \overline{A}_L^{[n_x]} \\ h^{[n_x-1, n_x]} \\ B^{[n_x-1]} A_R^{[n_x]} \end{pmatrix} + \begin{pmatrix} \overline{A}_L^{[n_x-1]} \overline{A}_L^{[n_x]} \\ h^{[n_x-1, n_x]} \\ A_L^{[n_x-1]} B^{[n_x]} \end{pmatrix} + \left(L_h^{[n_x-1]} \right) \begin{pmatrix} \overline{A}_L^{[n_x]} \\ B^{[n_x]} \end{pmatrix} + \left(L_{Bh}^{[n_x-1]} \right) \begin{pmatrix} \overline{A}_L^{[n_x]} \\ A_R^{[n_x]} \end{pmatrix} \\ \left(R_B^{[n_x]} \right) &= \begin{pmatrix} \overline{A}_R^{[n_x]} \\ B^{[n_x]} \end{pmatrix} + \begin{pmatrix} \overline{A}_R^{[n_x]} \\ A_L^{[n_x]} \end{pmatrix} \left(R_B^{[n_x+1]} \right) \end{aligned}$$

$$\partial_{\vec{B}^{(n_x, n_y)}} \langle \psi(\vec{B}) | H | \psi(B) \rangle = \sum$$



Conclusion

- The isometric structure of isoPEPS can be explicitly used to improve the boundary compression



- Parametrized tangent space vector provides faithful quasiparticle excitation ansatz also in 2D!

$$|\psi(X; A_L, C)\rangle = \sum_{n_x=1}^{2L_x} \sum_{y=1}^{L_y} \text{[Diagram 1]} + \sum_{n_y=1}^{2L_y-1} \text{[Diagram 2]}$$

$[n_x, y]$

$[n_y]$

$\overline{A}_L^{[n_x, y]}$
 $V_L^{[n_x, y]}$

$= 0$

$\overline{C}_D^{[n_y]}$
 $V_D^{[n_y]}$

$= 0$

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Thank you!



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